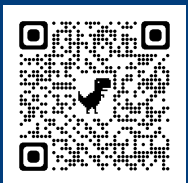


Towards Predictive Cities: Modeling Spatio-Temporal Data in the AI Era

Personal
Homepage



Xia Yutong

PhD Candidate

Institute of Data Science

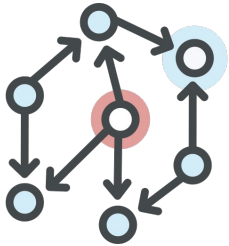
National University of Singapore





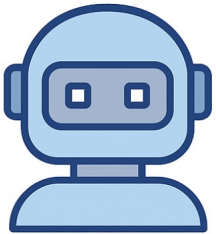
Background

- What makes cities predictable?
- Spatio-Temporal (ST) Data & Properties



Spatio-Temporal Graph Forecasting

- What is Spatio-Temporal Graph (STG)?
- What is STG forecasting?
- How we do it?
 - Application-Driven (Air Quality, Traffic, Parking)
 - Theory-Driven (Causality, Uncertainty)



Beyond Prediction: What's Next?

- LLMs-powered Agents & Causal Urban Insight



Background

- What makes cities predictable?
- Spatio-Temporal (ST) Data & Properties

Cities are not static structures. They are dynamic organisms — pulsing with people, data, and change.



Cities Are Alive — And Becoming Predictive

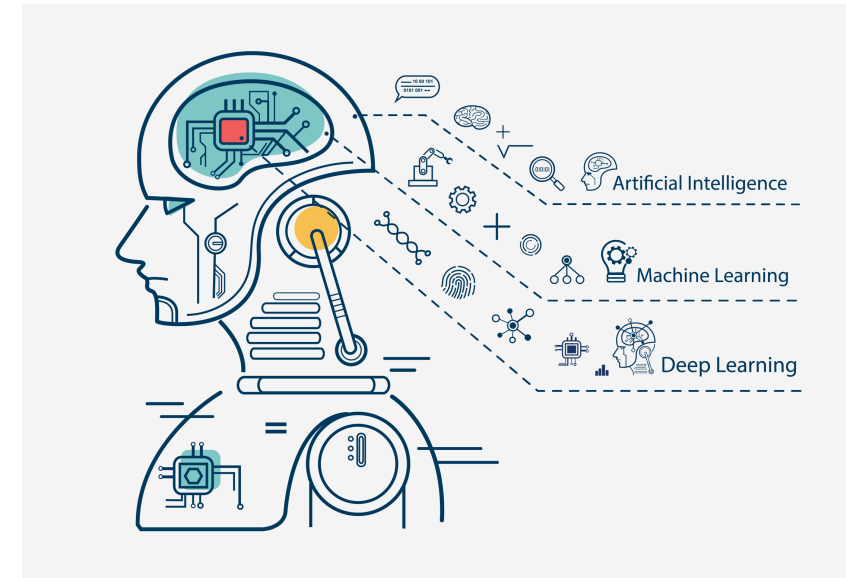
Nowadays...



Cities are **Data-Rich**

Recent advances in sensing technologies (e.g., IoT devices, mobile apps, satellite imagery, and urban sensors) have enabled the continuous collection of rich **spatio-temporal data**.

+



AI is a Powerful Tool

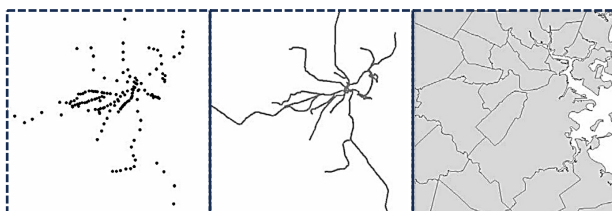
Advances in **AI (ML/DL/LLM)** provide tools to analyze complex patterns, forecast urban dynamics, and support data-driven decision-making.

Spatio-Temporal (ST) Data

- **Spatio-Temporal (ST) Data** is data that changes both over *space* and *time*. Tells us not just what is happening, but also *where* and *when* it's happening.
- Type of Urban ST Data

Spatio-Temporal Static Data (GeoData)

Vector

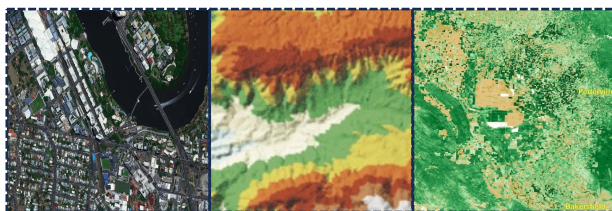


Points

Lines

Polygons

Raster

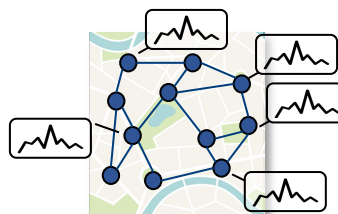


Satellite Images

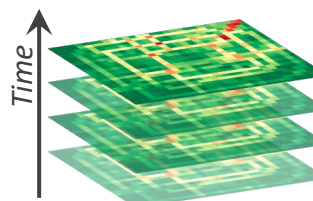
DEMs

NDVI

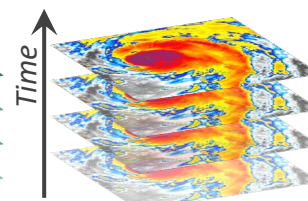
Spatial Static, Temporal Dynamic Data (Aggregate-level)



Weather/AQI/Traffic Sensor Data



OD Matrix



Gridded Weather Data

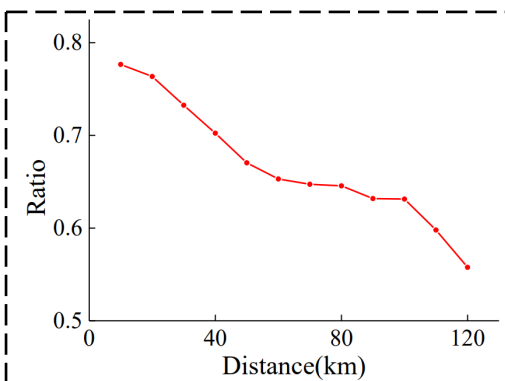
Spatio-Temporal Dynamic Data (Individual-level)



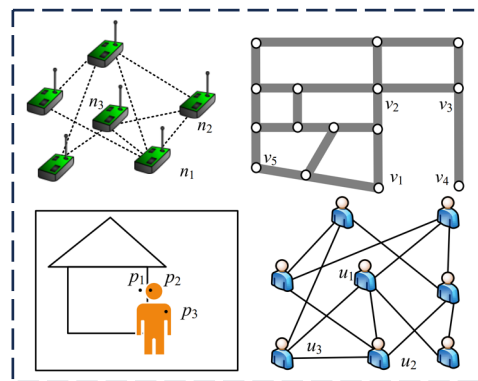
Trajectory Data

Spatial and Temporal Properties

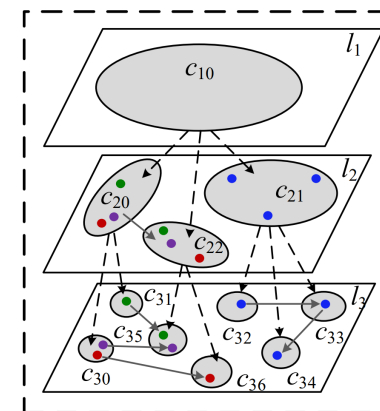
- Spatial Properties**



Spatial Closeness

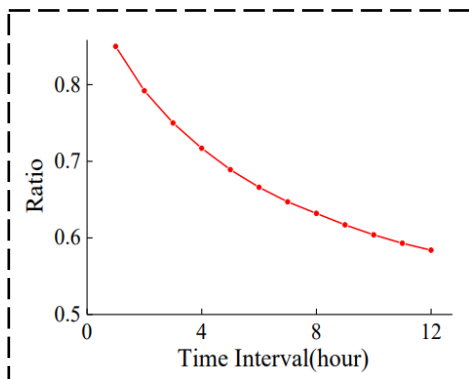


Spatial Distance

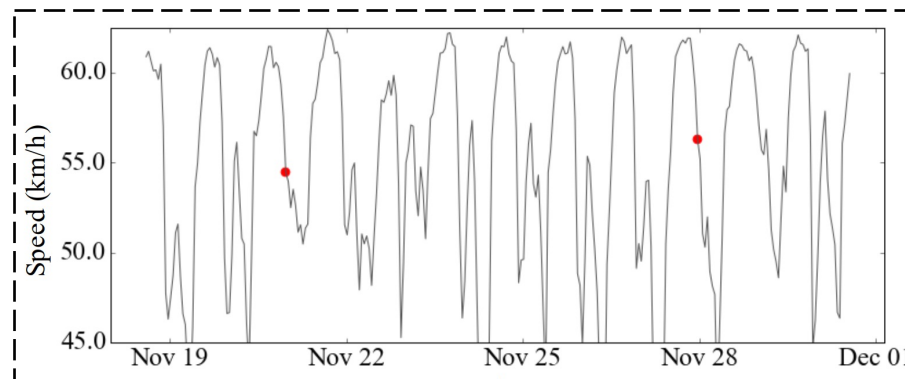


Spatial Hierarchy

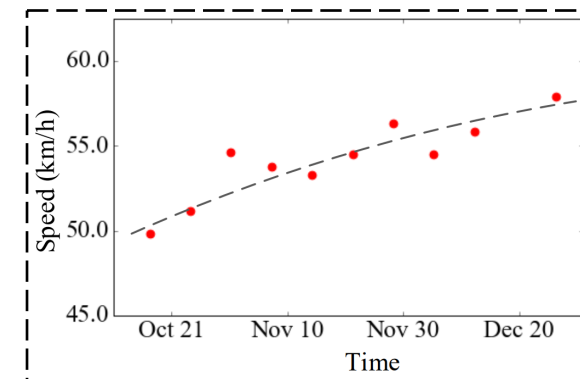
- Temporal Properties**



Temporal Closeness



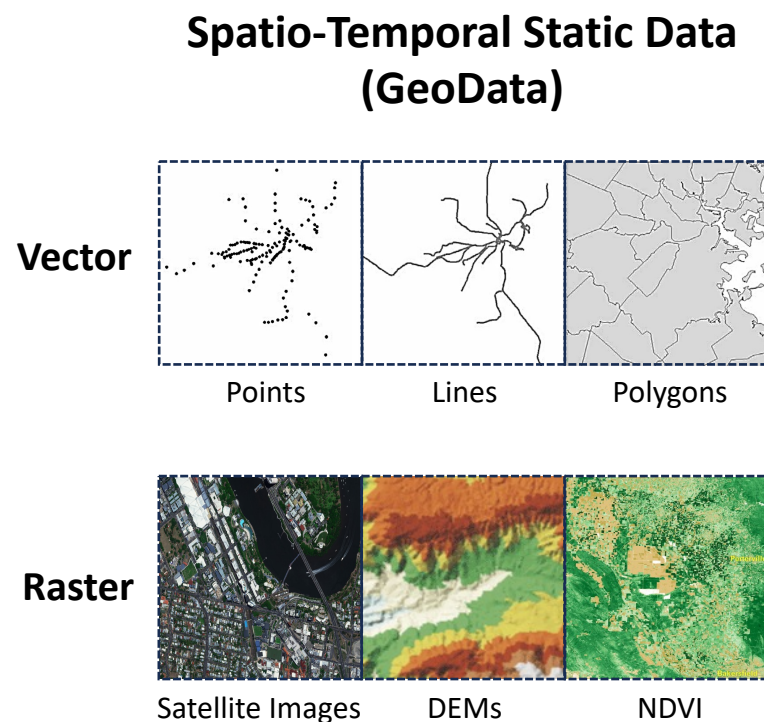
Temporal Periodicity



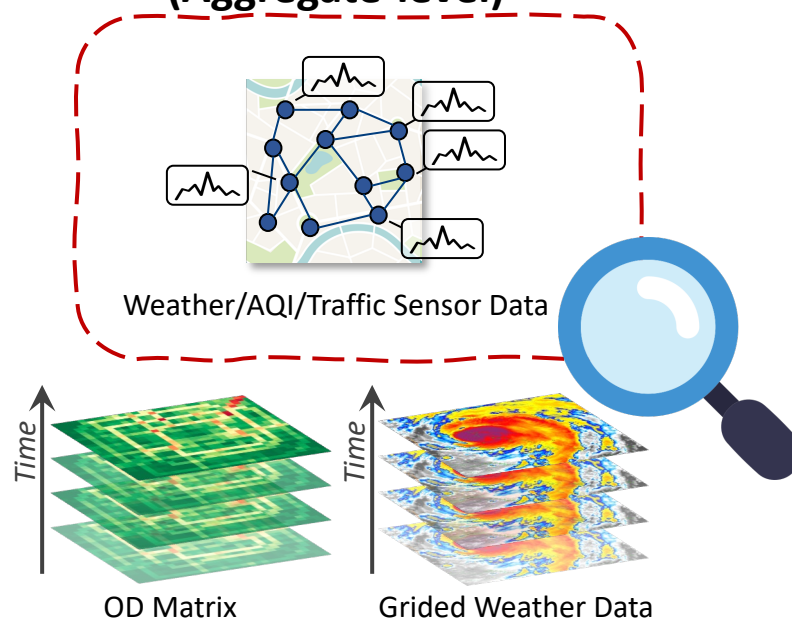
Trend

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Spatial Static, Temporal Dynamic Data (Aggregate-level)



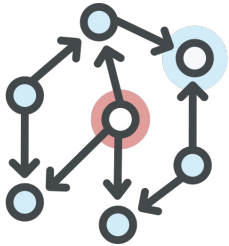
Spatio-Temporal Dynamic Data (Individual-level)





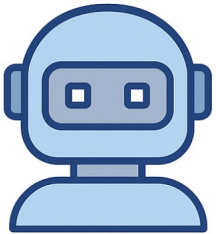
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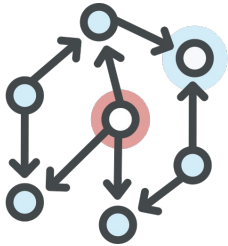
Spatio-Temporal Graph Forecasting

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Beyond Prediction: What's Next?

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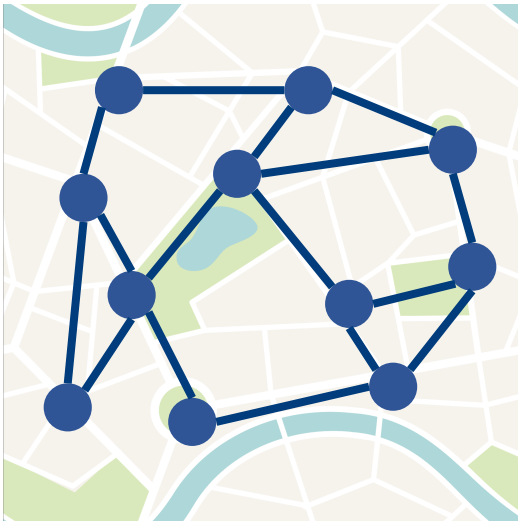


Spatio-Temporal Graph Forecasting

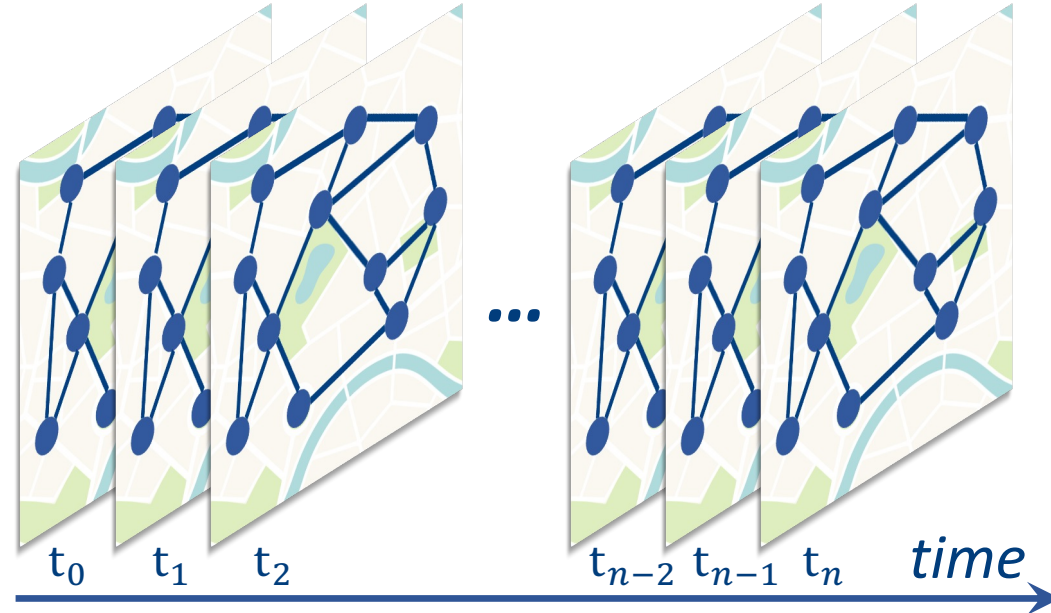
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Spatio-Temporal Graph (STG) Data

- **Spatio-Temporal Graph (STG)** is one type of ST data, which represents the spatial and temporal relationships between nodes or entities.



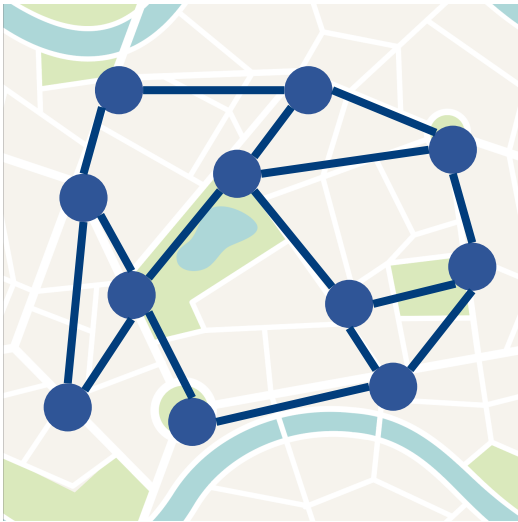
Graph



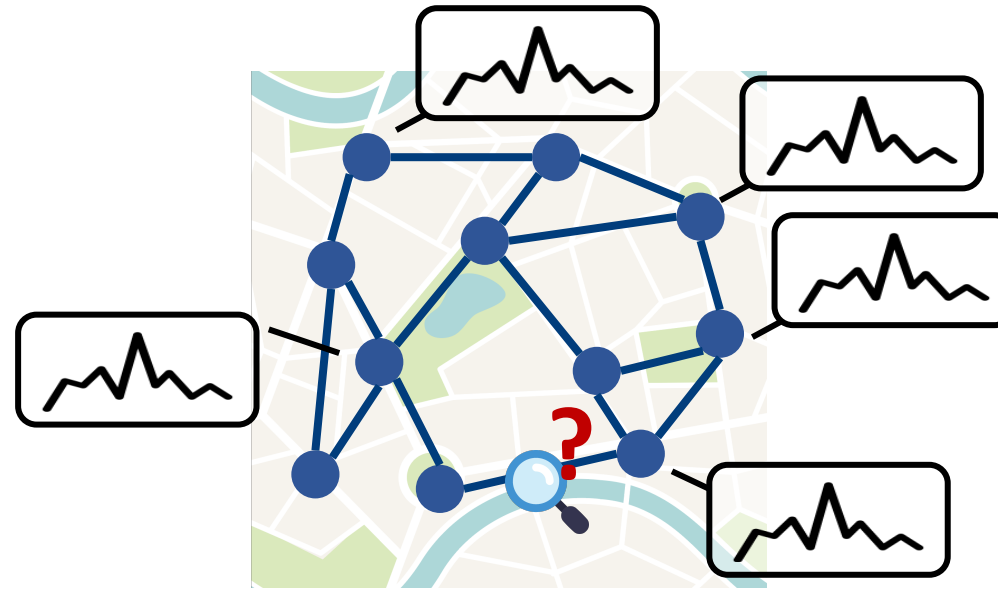
Spatio-Temporal Graph

Spatio-Temporal Graph (STG) Data

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Graph



Spatio-Temporal Graph (Series)

Spatio-Temporal Graph (STG) Data

- **Graph Construction Methods**

- Topology-based graph

$$a_{ij}^t = \begin{cases} 1, & \text{if } v_i \text{ connects to } v_j, \\ 0, & \text{otherwise} \end{cases},$$

- Distance-based graph

$$a_{ij}^t = \begin{cases} \frac{\exp(-\|d_{ij}^t\|_2)}{\sigma}, & \text{if } d_{ij}^t < \epsilon, \\ 0, & \text{otherwise} \end{cases}$$

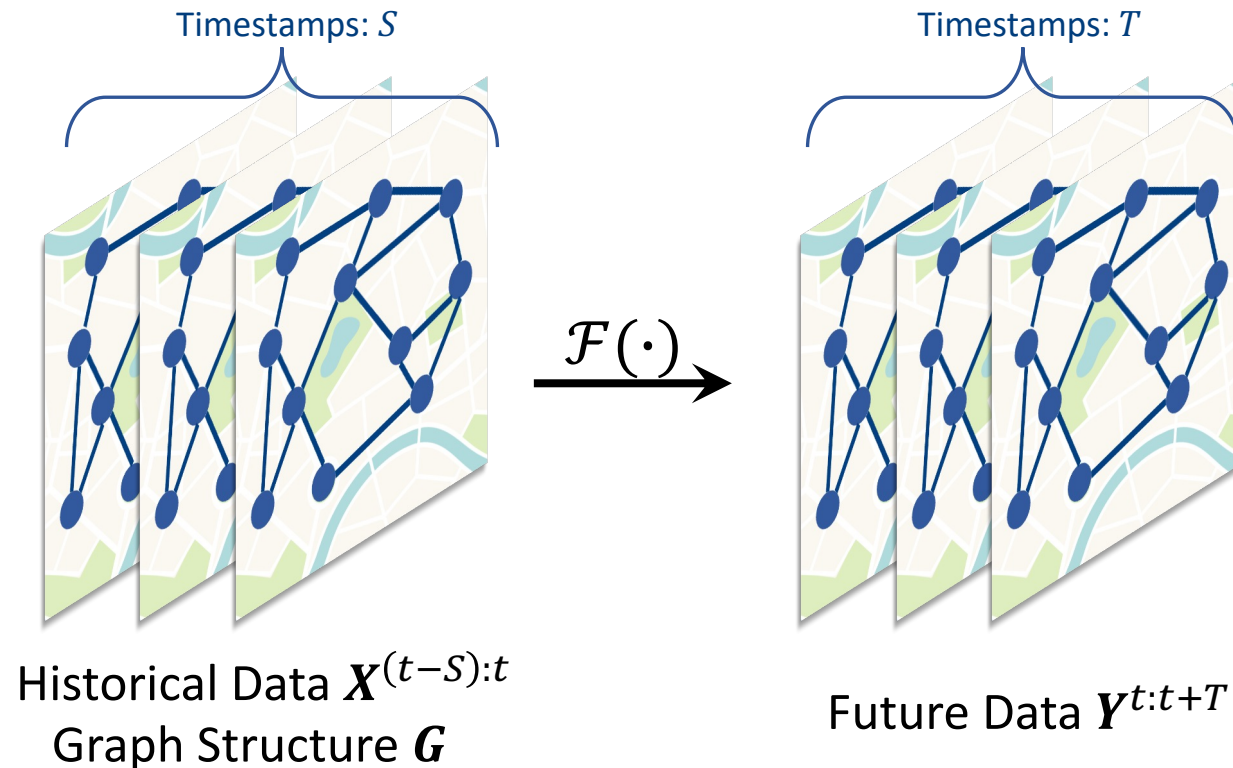
- Similarity-based graph

$$a_{ij}^t = \begin{cases} \frac{\sum_{i=1}^n (x_i^{0:t} - x_i^{\bar{0}:t}) (x_j^{0:t} - x_j^{\bar{0}:t})}{\sqrt{\sum_{i=1}^n (x_i^{0:t} - x_i^{\bar{0}:t})^2} \sqrt{\sum_{i=1}^n (x_j^{0:t} - x_j^{\bar{0}:t})^2}}, \\ 0, & \text{otherwise} \end{cases}$$

- Interaction-based graph

$$a_{ij}^t = \begin{cases} \frac{F_{ij}^t}{\sum_{m \in N(i)} F_{im}^t}, & \text{if } F_{ij}^t > 0, \\ 0, & \text{otherwise} \end{cases}$$

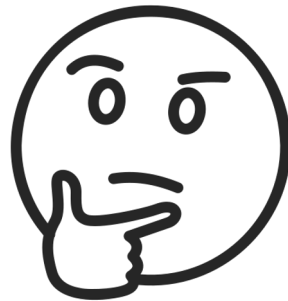
- **STG forecasting** has become crucial in the context of smart cities (e.g. Air quality prediction, traffic flow forecasting...)



How to learn $\mathcal{F}(\cdot)$?

Theory-Driven

- Causal Inference
- Uncertainty Awareness



Application-Driven

- Air Quality
- Traffic Flow
- Parking Availability



AirFormer (AAAI'23)

- **Challenges:** Inefficiency & Uncertainty
- **Solution:**

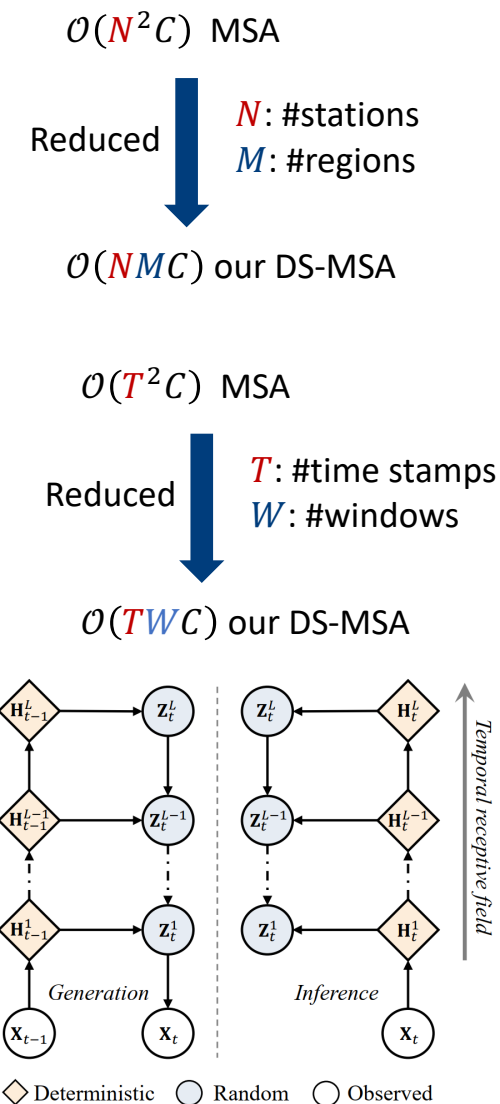
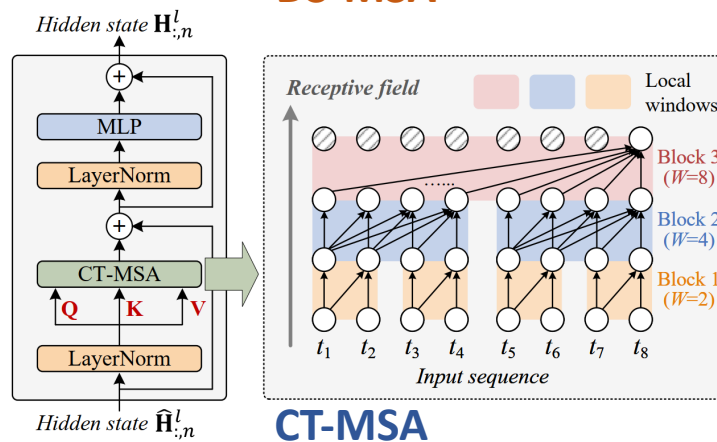
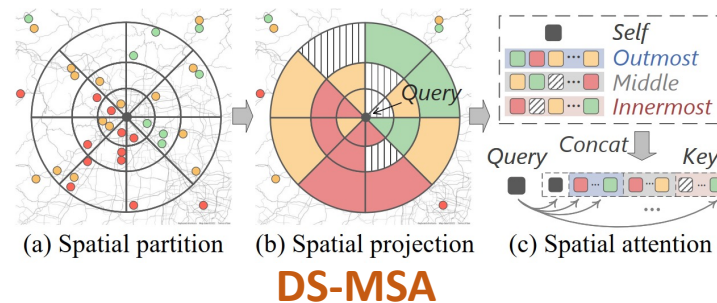
- Bottom-up deterministic stage
 - Dartboard Spatial-MSA (DS-MSA)
 - Causal Temporal-MSA (CT-MSA)

- Top-down stochastic stage
 - Generation model

$$\begin{aligned} p_{\theta}(\mathcal{Z}_t | \mathbf{X}_{1:t-1}) &= \prod_{n=1}^N p_{\theta}(\{\mathbf{z}_{t,n}^1, \dots, \mathbf{z}_{t,n}^L\} | \mathbf{X}_{1:t-1}) \\ &= \prod_{n=1}^N p_{\theta}(\mathbf{z}_{t,n}^L | \mathbf{h}_{t-1,n}^L) \prod_{l=1}^{L-1} p_{\theta}(\mathbf{z}_{t,n}^l | \mathbf{z}_{t,n}^{l+1}, \mathbf{h}_{t-1,n}^l), \end{aligned}$$

- Inference model

$$q_\phi(\mathcal{Z}_t|\mathbf{X}_{1:t}) = \prod_{n=1}^N q_\phi(\mathbf{z}_{t,n}^L|\mathbf{h}_{t,n}^L) \prod_{l=1}^{L-1} q_\phi(\mathbf{z}_{t,n}^l|\mathbf{z}_{t,n}^{l+1}, \mathbf{h}_{t,n}^l)$$



Application-Driven Method

Air Quality Forecasting



Traffic Flow Forecasting

Parking Availability Forecasting

LargeST (NeurIPS'23)

Motivation – Limitation in Existing Dataset

Scale: Existing datasets (e.g., PeMS03, PeMS04) contain only hundreds of sensors, not reflecting real-world traffic network scales.

Temporal Coverage: Typically cover less than 6 months, hindering the study of long-term patterns.

Metadata: Often lack comprehensive sensor metadata, affecting data reliability and interpretability



LargeST - A new large-scale dataset

Larger Graph Size: 8,600 sensors across California.

Higher Temporal Coverage: 5 years of data (2015–2019) with a 5-minute sampling rate.

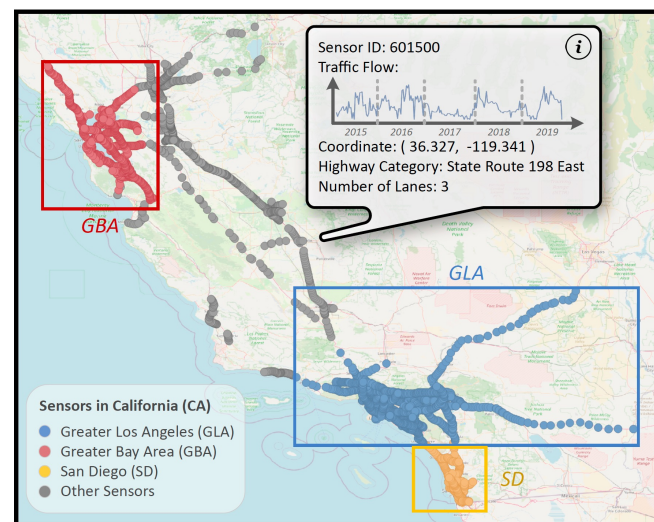
Richer Node Metadata: Includes sensor ID, location, highway category, number of lanes, and direction.

Subsets: Provides regional subsets for Greater Los Angeles (GLA), Greater Bay Area (GBA), and San Diego (SD).

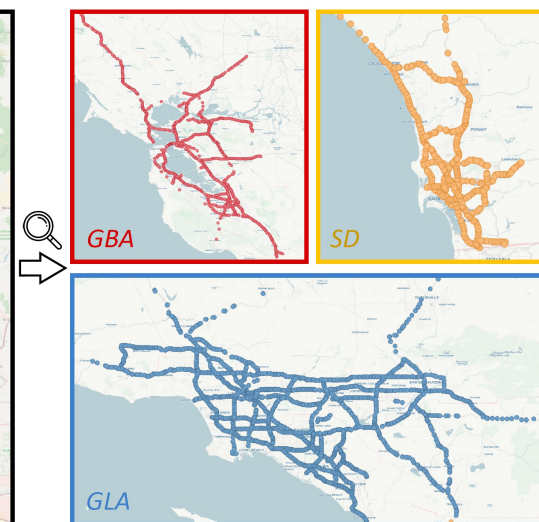


**Dataset
& Code**

Released Source	Dataset	Nodes	Edges	Degree	Meta	Time Range	Frames	Data Points
Yu et al. [34]	PeMSD7(M)	228	1,664	7.3	6	05/01/2012 – 06/30/2012	12,672	2.89M
	PeMSD7(L)	1,026	14,534	14.2	0	05/01/2012 – 06/30/2012	12,672	13.00M
Li et al. [19]	METR-LA	207	1,515	7.3	3	03/01/2012 – 06/27/2012	34,272	7.09M
	PEMS-BAY	325	2,369	7.3	3	01/01/2017 – 06/30/2017	52,116	16.94M
Song et al. [30]	PEMS03	358	546	1.5	1	09/01/2018 – 11/30/2018	26,208	9.38M
	PEMS04	307	338	1.1	0	01/01/2018 – 02/28/2018	16,992	5.22M
	PEMS07	883	865	1.0	0	05/01/2017 – 08/06/2017	28,224	24.92M
	PEMS08	170	276	1.6	0	07/01/2016 – 08/31/2016	17,856	3.04M
LargeST (ours)	CA	8,600	201,363	23.4	9	01/01/2017 – 12/31/2021	525,888	4.52B
	GLA	3,834	98,703	25.7	9	01/01/2017 – 12/31/2021	525,888	2.02B
	GBA	2,352	61,246	26.0	9	01/01/2017 – 12/31/2021	525,888	1.24B
	SD	716	17,319	24.2	9	01/01/2017 – 12/31/2021	525,888	0.38B



(a) Overview of the LargeST dataset



(b) Fine-grained distribution of sensors

Application-Driven Method

Air Quality Forecasting

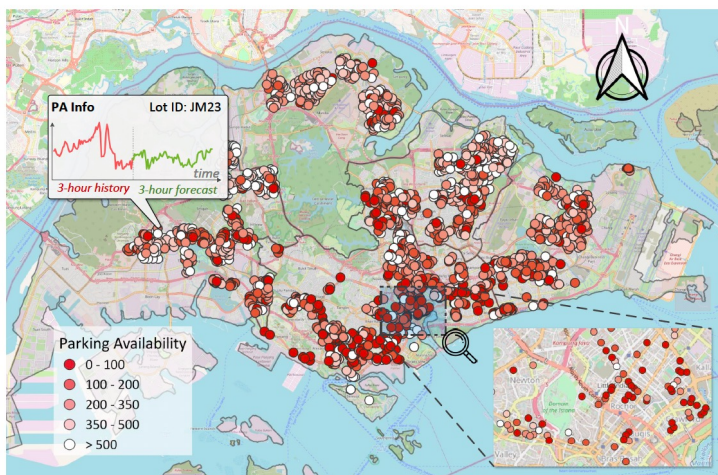
Traffic Flow Forecasting



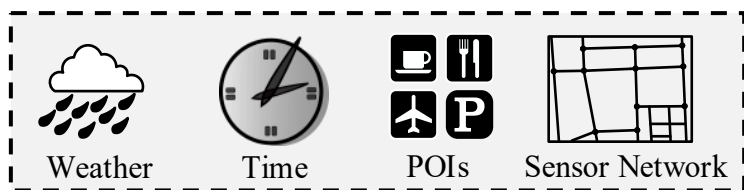
Parking Availability Forecasting

DeepPA + SINPA (IJCAI'24)

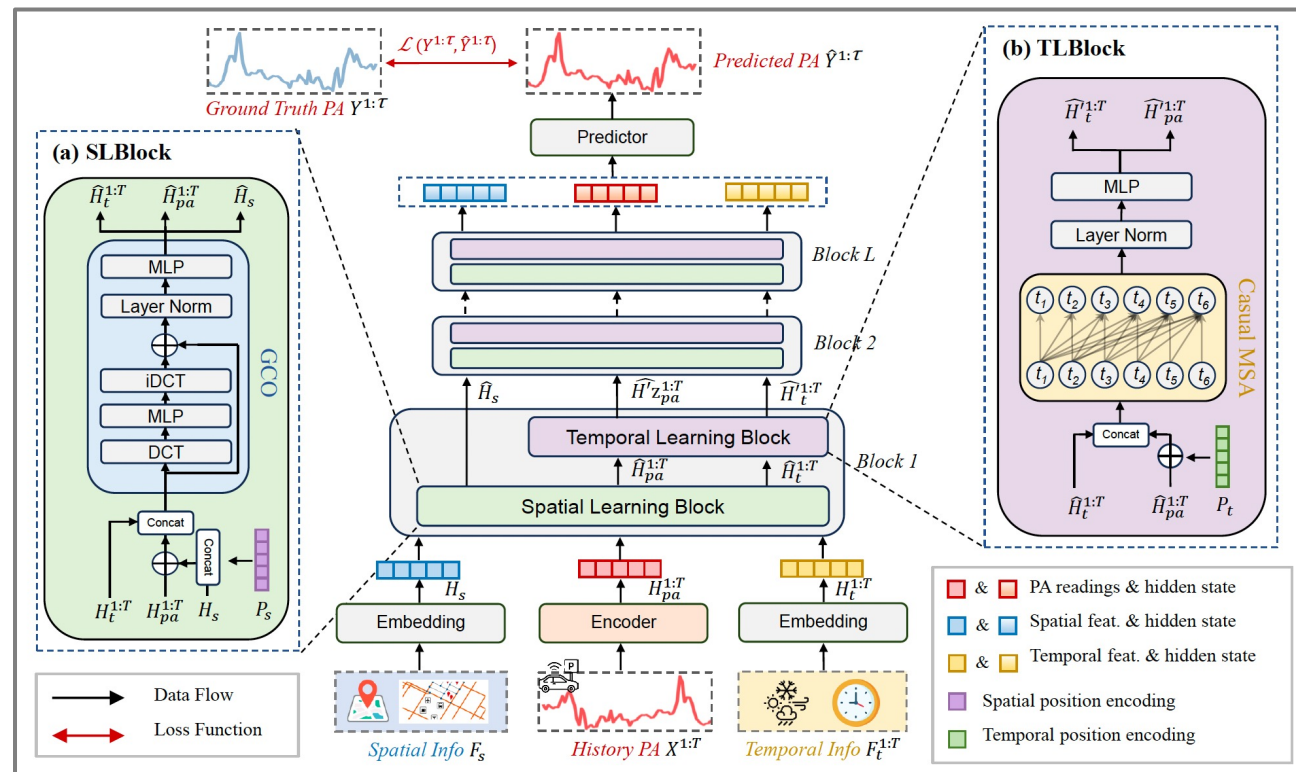
Predicting Parking Availability in Singapore with Cross-Domain Data



Parking Availability Data



Cross-domain Data



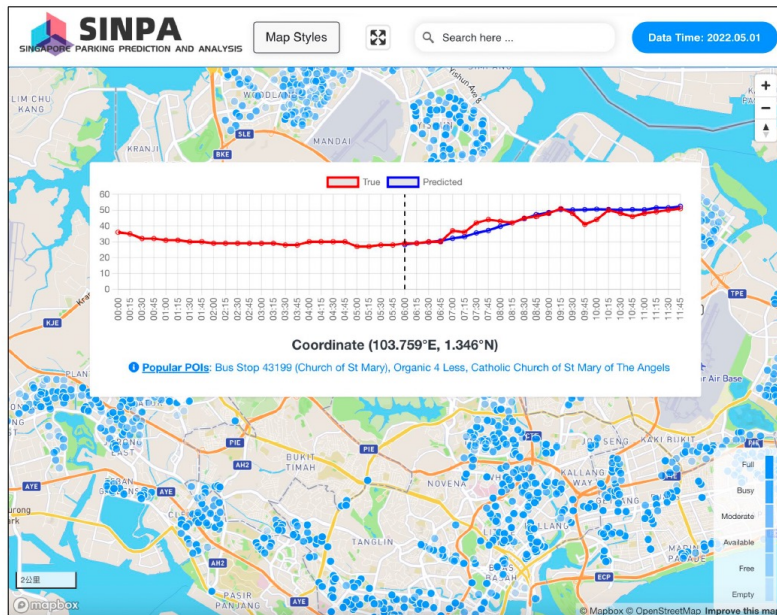
A New Dataset + A Novel Data-driven Method



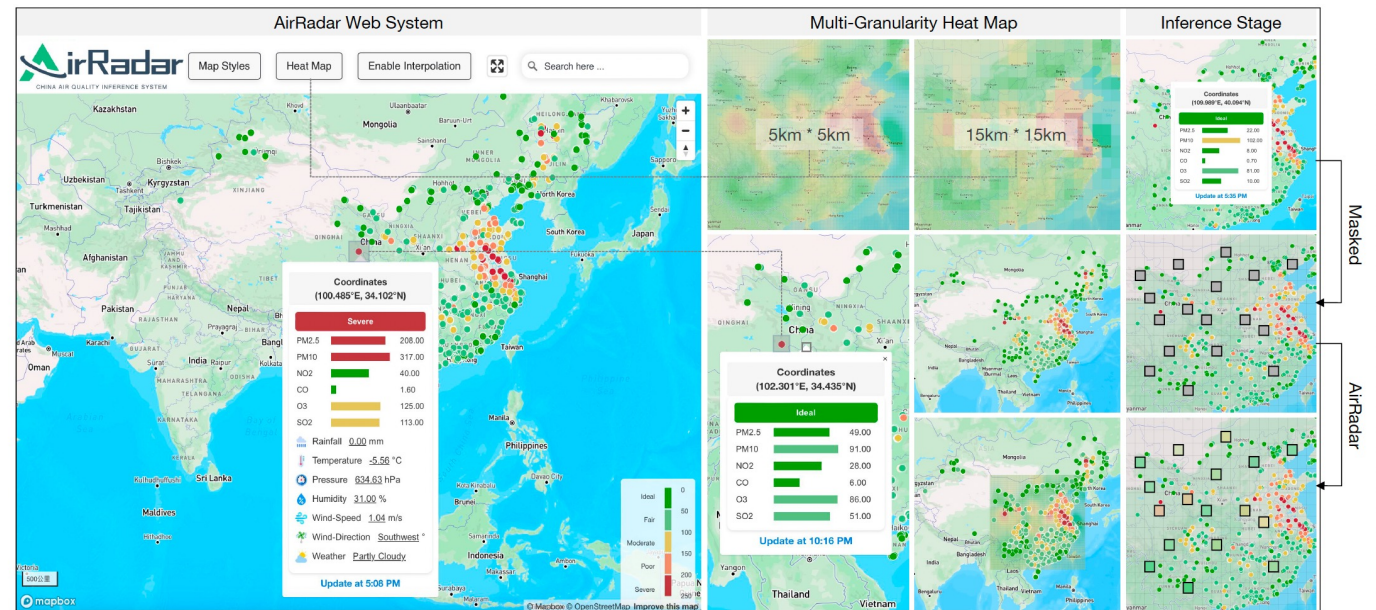
Dataset & Code

Application-Driven Method

For application, we also **deploy** our cutting-edge AI solutions on large-scale cloud platforms



Parking Availability Prediction in Singapore

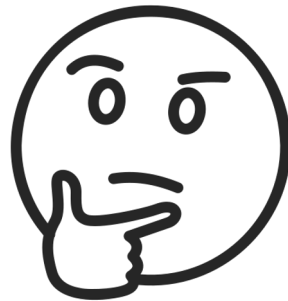


Air Quality Inference in China

How to learn $\mathcal{F}(\cdot)$?

Theory-Driven

- Causal Inference
- Uncertainty Awareness



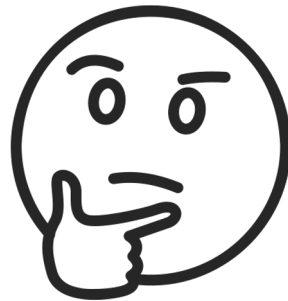
Application-Driven

- Air Quality
- Traffic Flow
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How to learn $\mathcal{F}(\cdot)$?

Theory-Driven

- **Causal Inference**
- Uncertainty Awareness



Application-Driven

- Air Quality
- Traffic Flow
- Parking Availability

Theory-Driven - *Causal Inference*

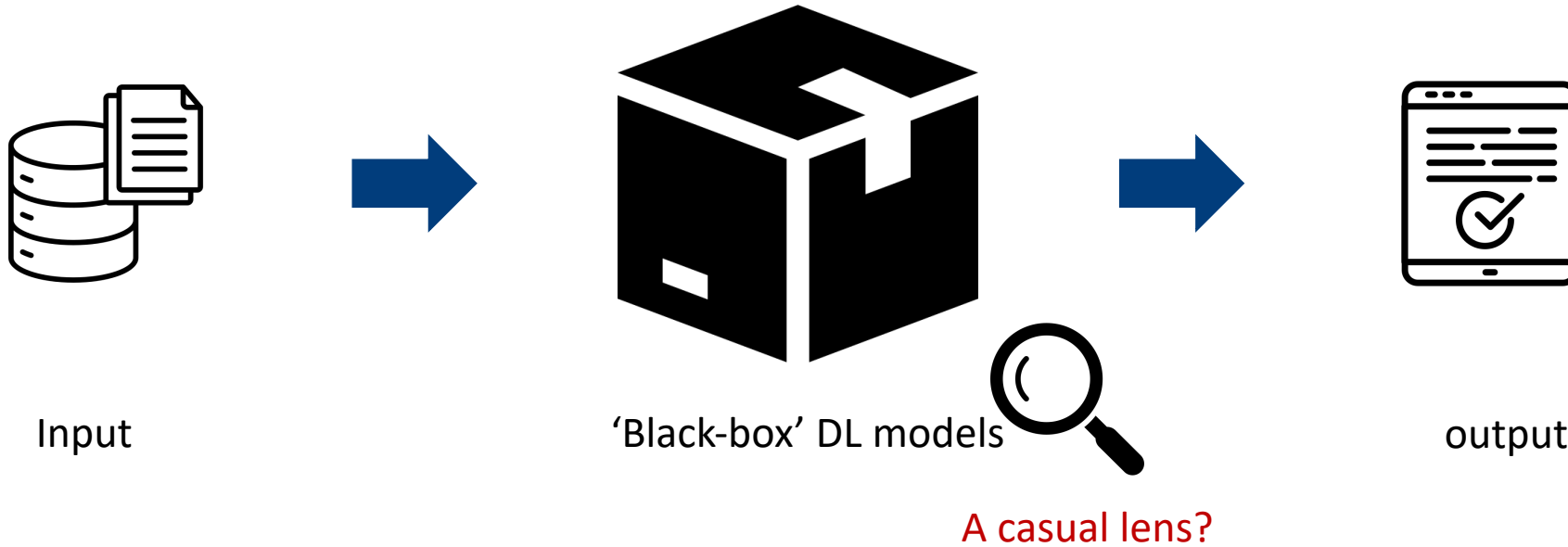
Motivation

A Causal Lens on STG Forecasting

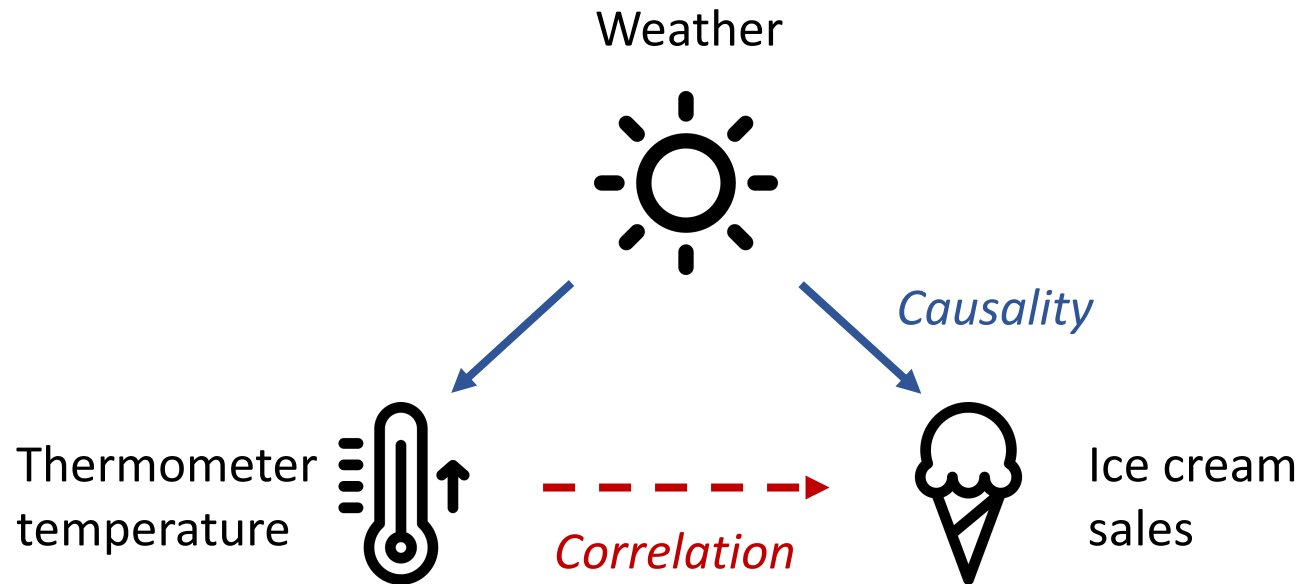
Causal Tools

A DL Implement

Experiments Results



- **Causal Inference** is the process of determining if a relationship between two things is a cause-and-effect relationship.
- *Correlation doesn't mean causality.*

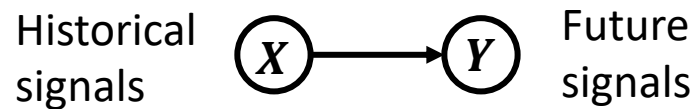


Integrating deep learning with causal inference, we craft models that are not only accurate but truly understand the real world.

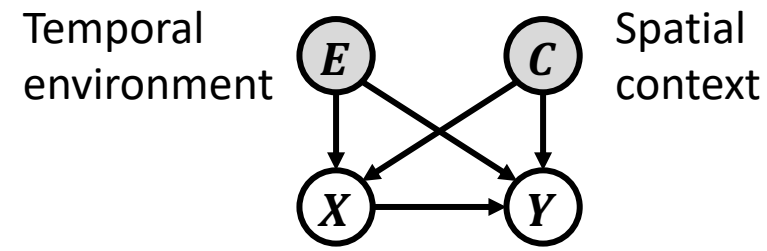
- **Why?** - Advantage of a causal lens
 - Improved Interpretability
 - Real-world insights for better model design
 - Enhanced generalization
- **How?** – A big picture



Structure Causal Model



Traditional model



Underlying causal system

Theory-Driven - Causal Inference

Motivation

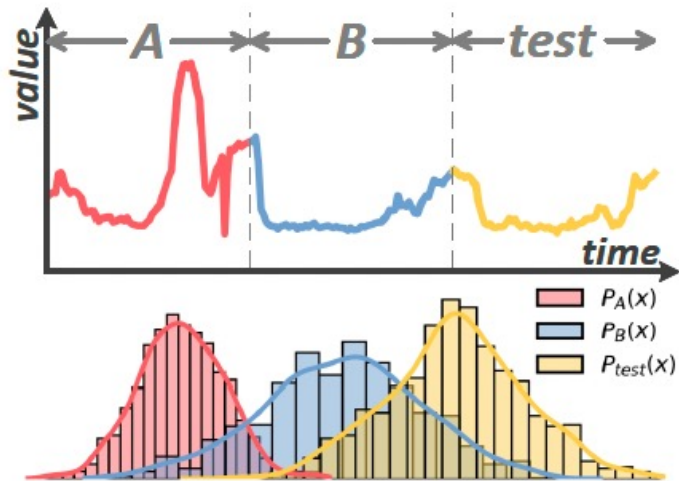
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Causal Tools

A DL Implement

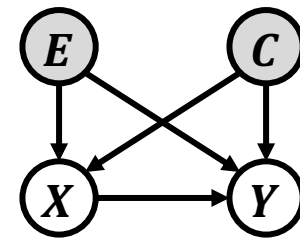
Experiments Results

Structure Causal Model



$$P_A(X) \neq P_B(X) \neq P_{test}(X)$$

Temporal
environment



Spatial
context

Underlying causal system

- $X \leftarrow E \rightarrow Y$: The **temporal Out-of-Distribution (OoD)** can arise due to changes in external variables over time (e.g., weather can affect traffic flow observations).
- $X \leftarrow C \rightarrow Y$: X and Y are intrinsically affected by the surrounding spatial context, comprising both **spurious** and **genuine** causal components.
- $X \rightarrow Y$: Our primary prediction goal.

Theory-Driven - Causal Inference

Motivation

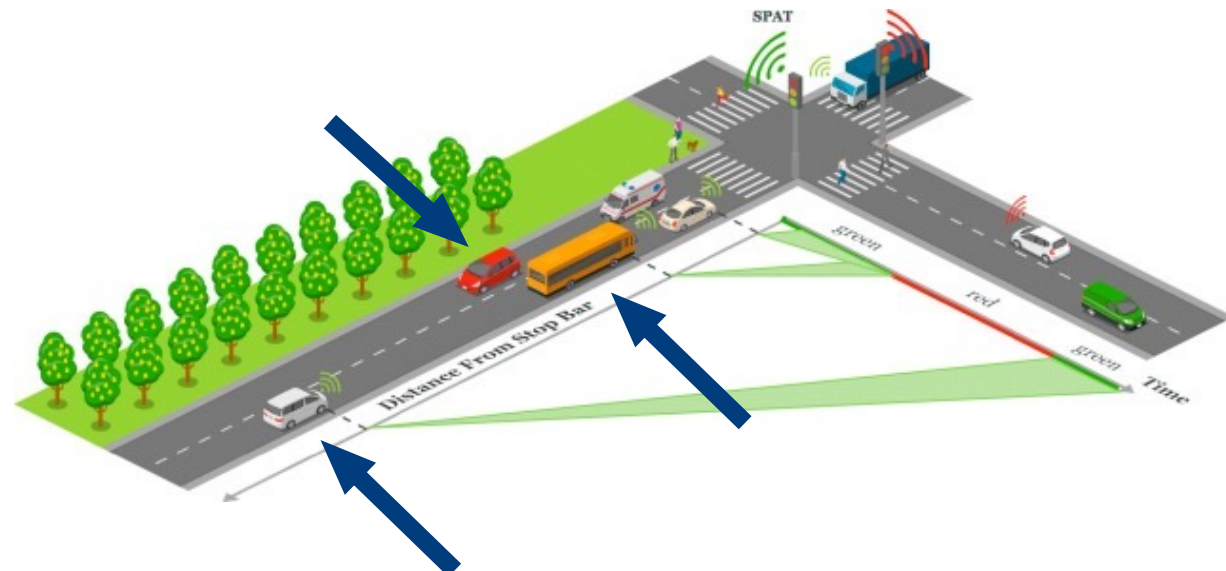
A Causal Lens on STG Forecasting

Causal Tools

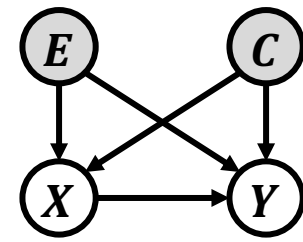
A DL Implement

Experiments Results

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Theory-Driven - Causal Inference

Motivation

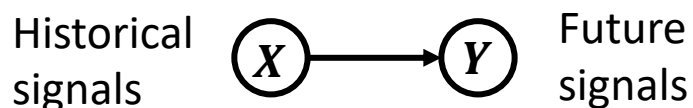
A Causal Lens on STG Forecasting

Causal Tools

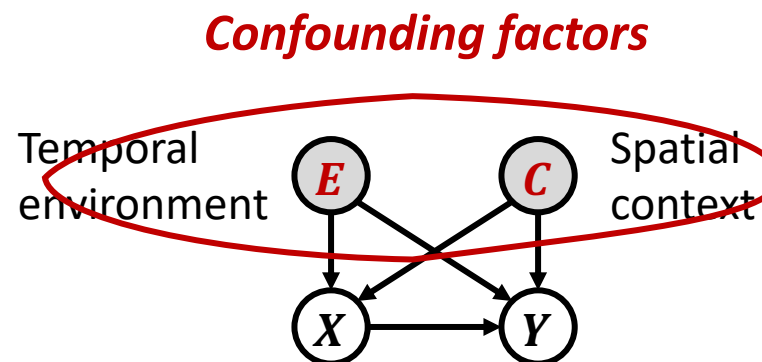
A DL Implement

Experiments Results

Structure Causal Model



Traditional model



Underlying causal system

Backdoor paths

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- $X \leftarrow C \rightarrow Y$: X and Y are intrinsically affected by the surrounding spatial context, comprising both **spurious** and **genuine** causal components.
- $X \rightarrow Y$: Our primary goal.

Assumption: E and C are independent

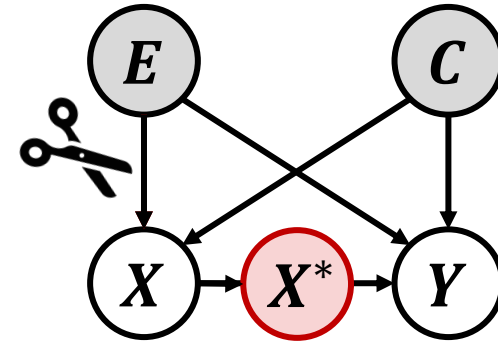
Aim: $P(Y|do(X))$

Step 1. Back-door adjustment for E

$$\sum_e P(Y|X, E = e)P(E = e)$$

Step 2. Front-door adjustment for C

$$\sum_{x^*} \sum_{x'} P(X^* = x^* | X) P(Y | X^* = x^*, X = x') P(X = x')$$

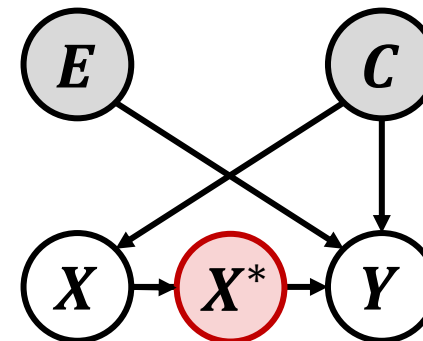


- To achieve **Back-door adjustment** ...

$$\sum_e P(Y|X, E = e)P(E = e)$$

- 1) Separate the entity and the environments
- 2) Discretizing the environment

- To achieve **Front-door adjustment** ...
 - Obtain a surrogate entity



Theory-Driven - Causal Inference

Motivation

A Causal Lens on STG Forecasting

Causal Tools

A DL Implement

Experiments Results

Causal Spatio-Temporal neural network (CaST)

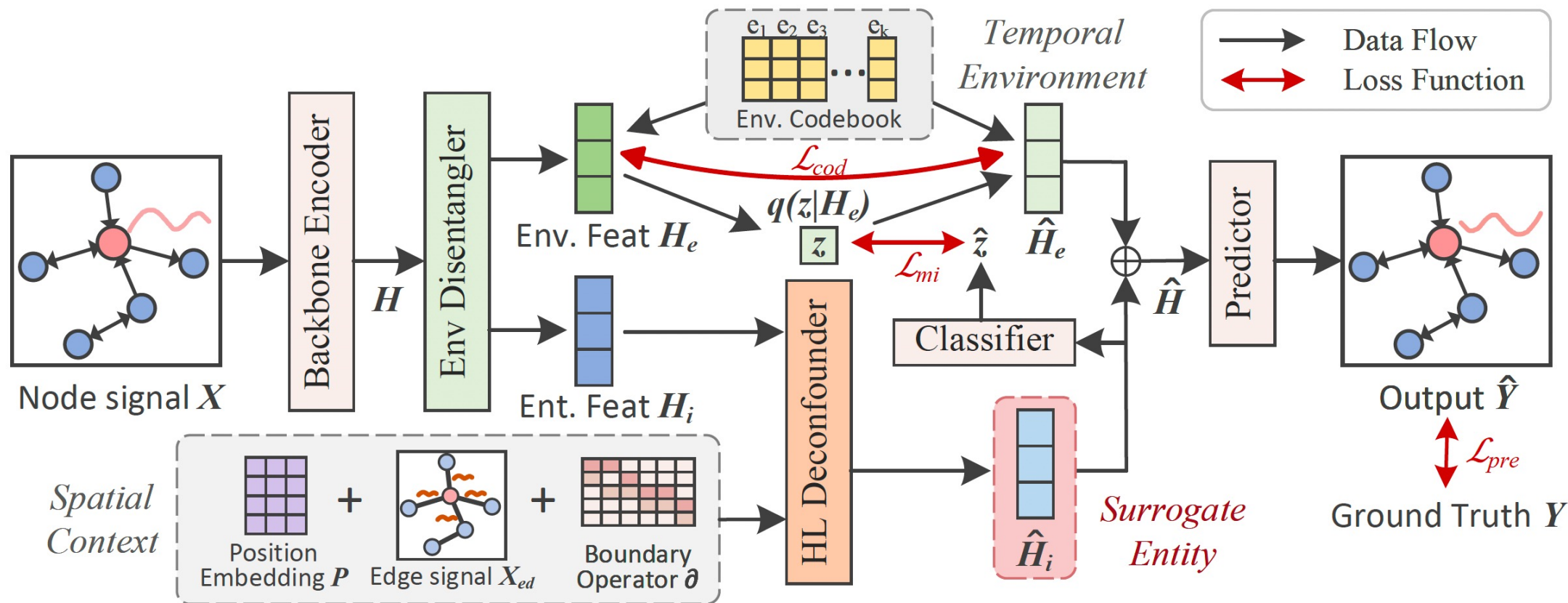


Figure 3: The pipeline of CaST. Env: Environment. Ent: Entity. Feat: Feature.

Back-door adjustment



Theory-Driven - Causal Inference

Motivation

A Causal Lens on STG Forecasting

Causal Tools

A DL Implement

Experiments Results

Causal Spatio-Temporal neural network (CaST)

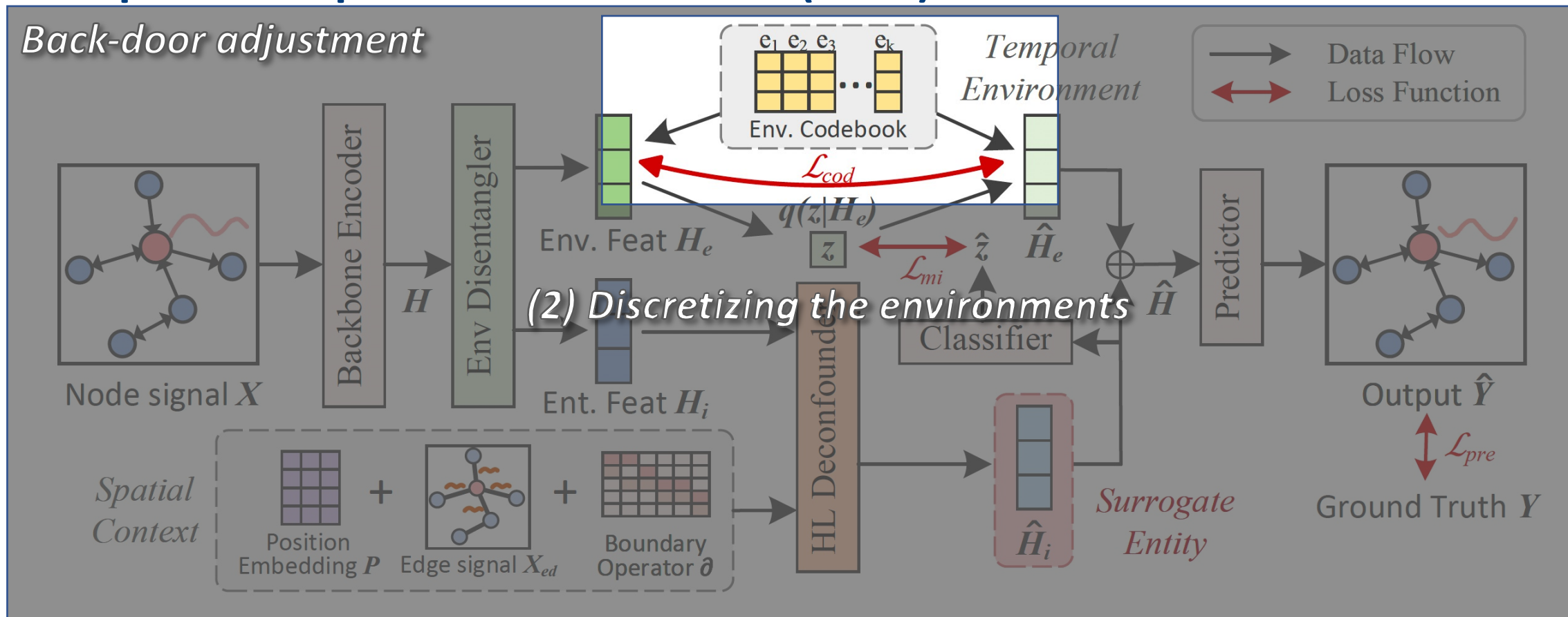


Figure 3: The pipeline of CaST. Env: Environment. Ent: Entity. Feat: Feature.

Theory-Driven - Causal Inference

Motivation

A Causal Lens on STG Forecasting

Causal Tools

A DL Implement

Experiments Results

Causal Spatio-Temporal neural network (CaST)

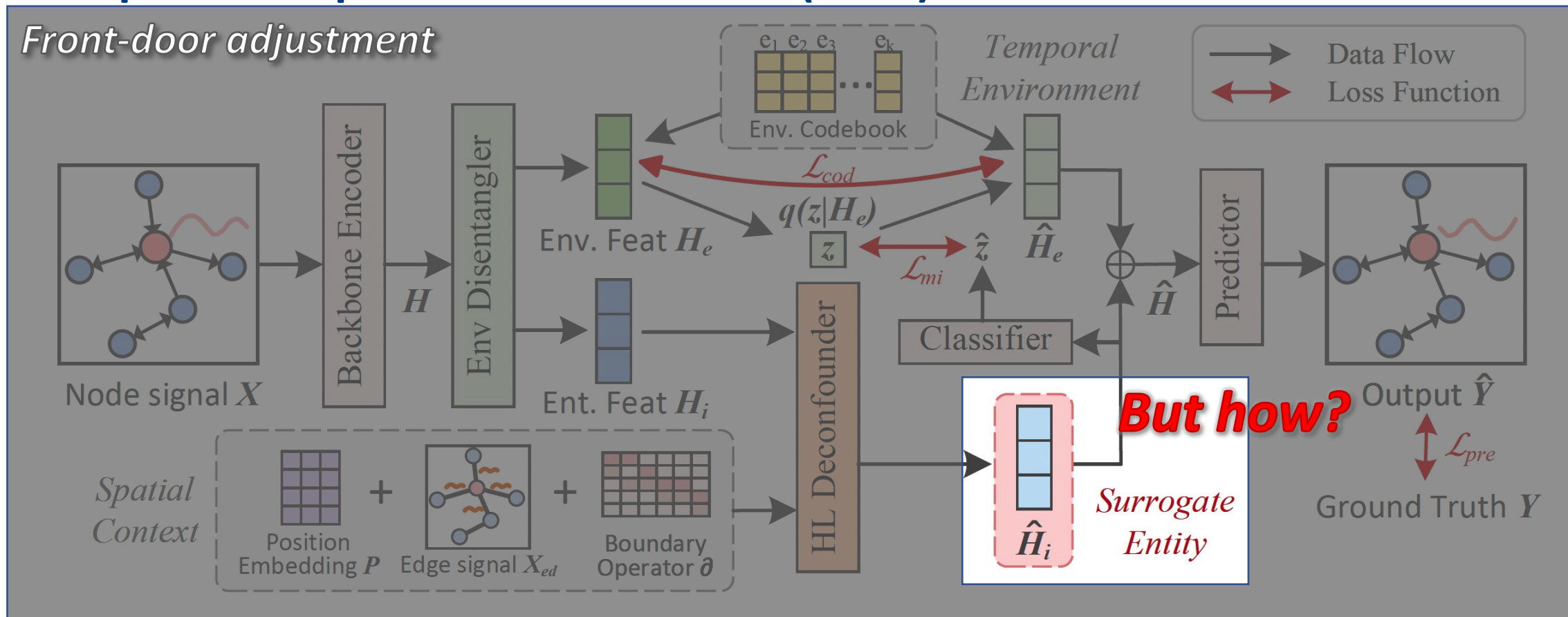


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- To achieve **Back-door adjustment** ...

$$\sum_e P(Y|X, E = e)P(E = e)$$

- 1) Separate the entity and the environments
- 2) Discretizing the environment

- To achieve **Front-door adjustment** ...
 - Obtain a **surrogate entity** – **But how?**
 - **Data/task-specific challenge** – Causation's ripple effect
 - Solution – Graph convolution networks?

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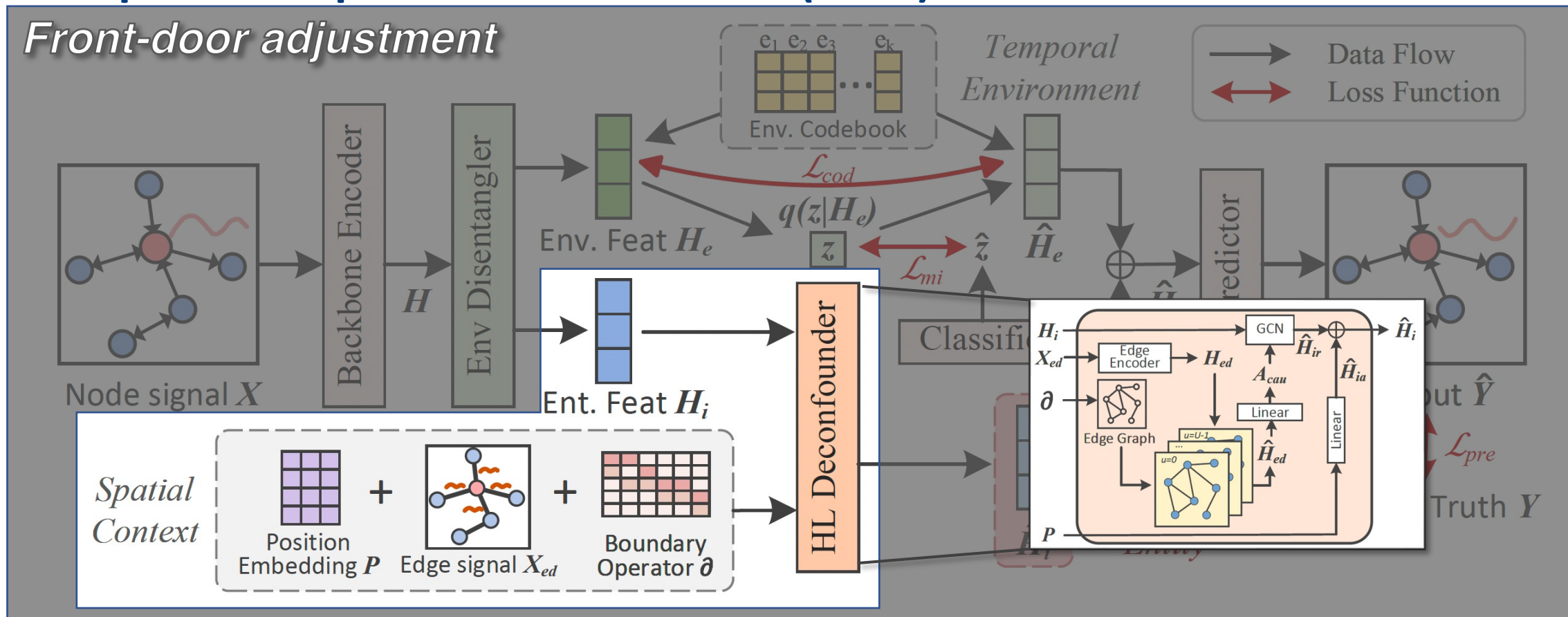


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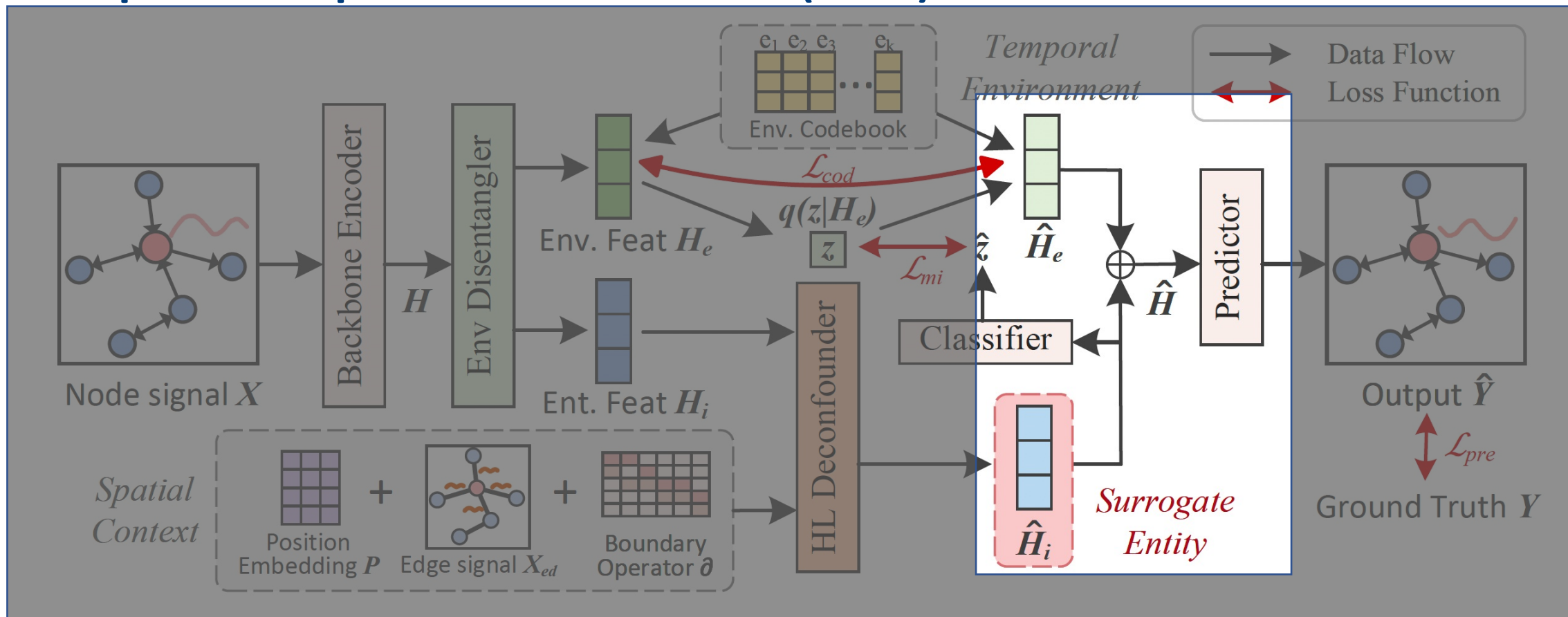


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Theory-Driven - Causal Inference

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A DL Implement

Experiments Results

- Datasets: PEMS08, AIR-BJ, AIR-GZ
- Experiment settings: predict over next 24 steps given past 24 steps
- Evaluation metrics: MAE, RMSE

Effectiveness & Generalizability

Interpretability

Table 1: 5-run error comparison. The bold/underlined font means the best/the second-best result.

Model	PEMS08 (24→24)		AIR-BJ (24→24)		AIR-GZ (24→24)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA(2017)	58.83	81.96	32.12	43.95	19.56	25.77
VAR(1991)	37.04	53.08	29.79	42.04	14.97	20.61
DCRNN(2017)	22.10 ± 0.45	33.96 ± 0.59	23.72 ± 0.36	35.84 ± 0.56	12.99 ± 0.26	18.27 ± 0.41
STGCN(2018)	18.60 ± 0.08	28.44 ± 0.15	23.71 ± 0.21	36.30 ± 0.58	12.69 ± 0.04	17.66 ± 0.09
ASTGCN(2019)	20.36 ± 0.48	30.87 ± 0.55	23.78 ± 0.22	35.91 ± 0.11	12.91 ± 0.15	18.02 ± 0.27
MTGNN(2020)	18.13 ± 0.10	28.85 ± 0.12	24.35 ± 0.74	38.97 ± 1.81	<u>12.43</u> ± 0.11	17.99 ± 0.18
AGCRN(2020)	<u>17.06</u> ± 0.14	<u>26.80</u> ± 0.15	<u>23.43</u> ± 0.29	<u>35.66</u> ± 0.57	12.74 ± 0.01	<u>17.49</u> ± 0.01
GMSDR(2022)	18.34 ± 0.68	28.36 ± 1.01	25.92 ± 0.52	39.60 ± 0.44	13.47 ± 0.31	19.04 ± 0.46
STGNCDE(2022)	17.55 ± 0.30	27.28 ± 0.36	24.35 ± 0.31	35.91 ± 0.48	13.70 ± 0.10	19.15 ± 0.07
CaST (ours)	16.44 ± 0.10	26.61 ± 0.15	22.90 ± 0.09	34.84 ± 0.11	12.36 ± 0.01	17.25 ± 0.05

Traffic Flow

Air Quality

Theory-Driven - Causal Inference

Motivation

A Causal Lens on STG Forecasting

Causal Tools

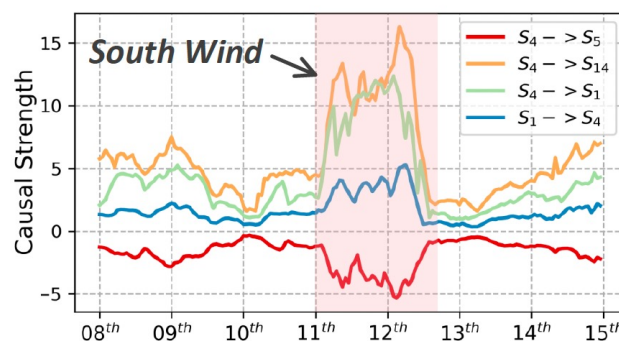
A DL Implement

Experiments Results

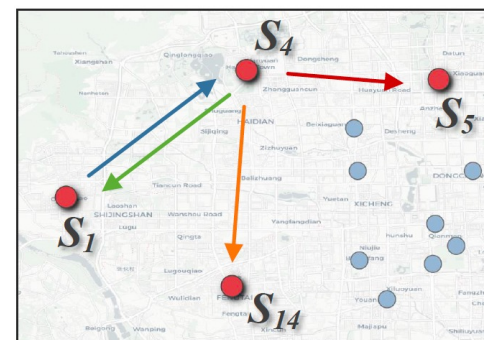
Dynamic Spatial Causation

Effectiveness & Generalizability

Interpretability

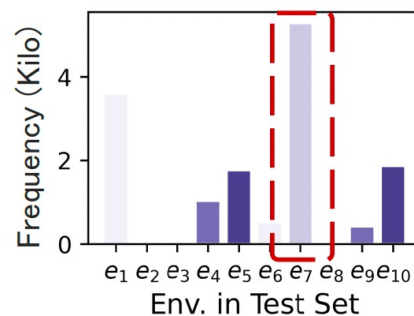
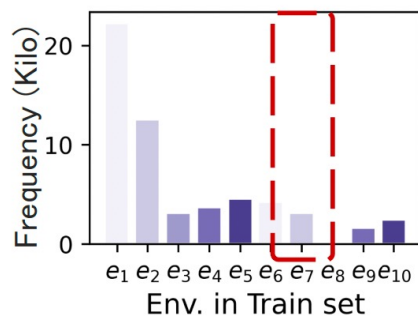


(b) Dynamic Causal Relations

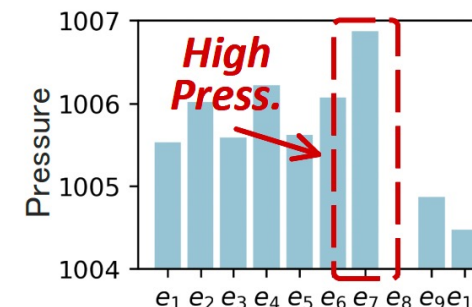
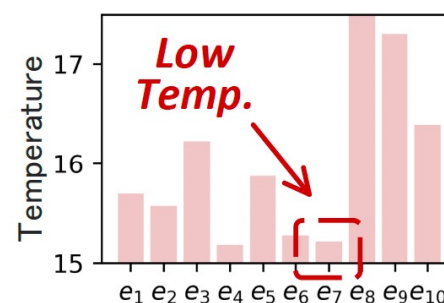


(c) Geographical Distribution

Temporal Environments



(b) Frequency of Environments

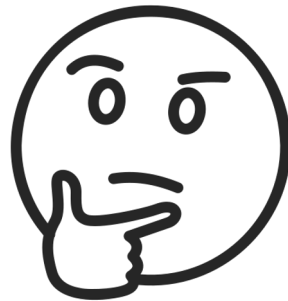


(c) External Factors of Environments

How to learn $\mathcal{F}(\cdot)$?

Theory-Driven

- Causal Inference
- **Uncertainty Awareness**



Application-Driven

- Air Quality
- Traffic Flow
- Parking Availability



Theory-Driven - *Uncertainty*

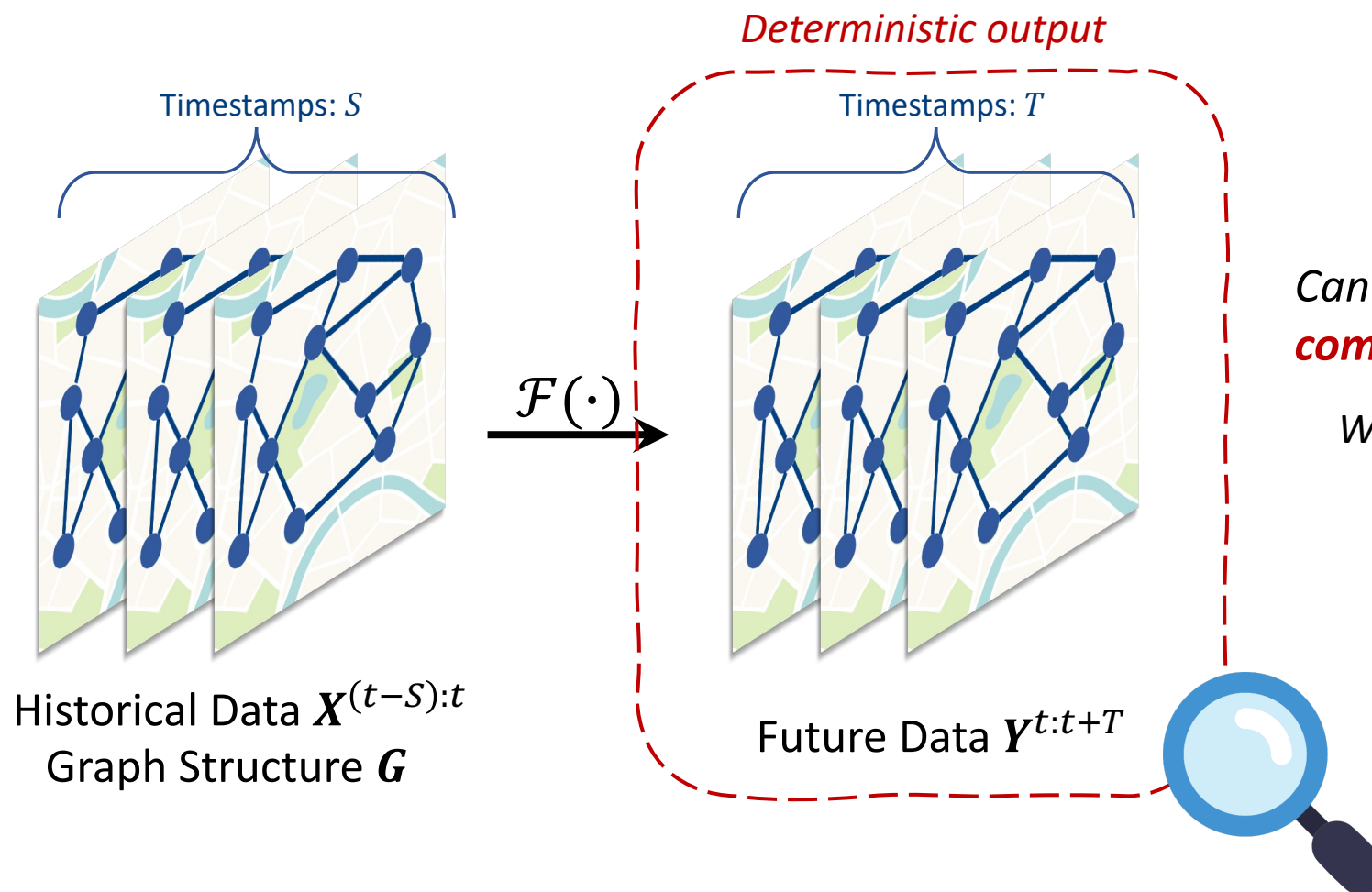
Motivation

Background – DDPM

Key Challenges

Our Solution - DiffSTG

Experiments Results

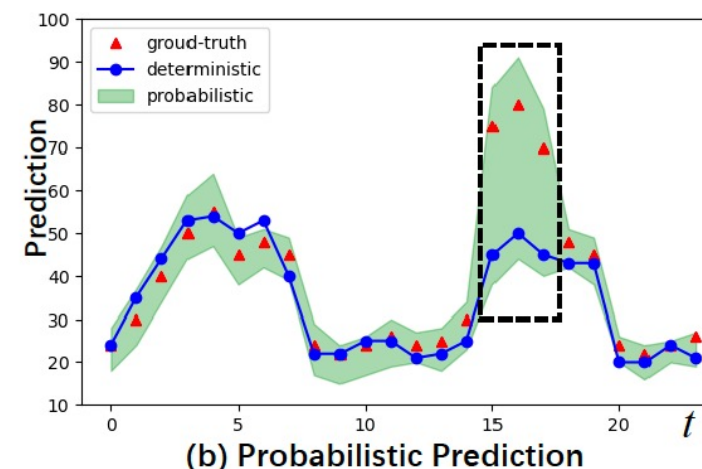


Urban dynamics are inherently **uncertain**.

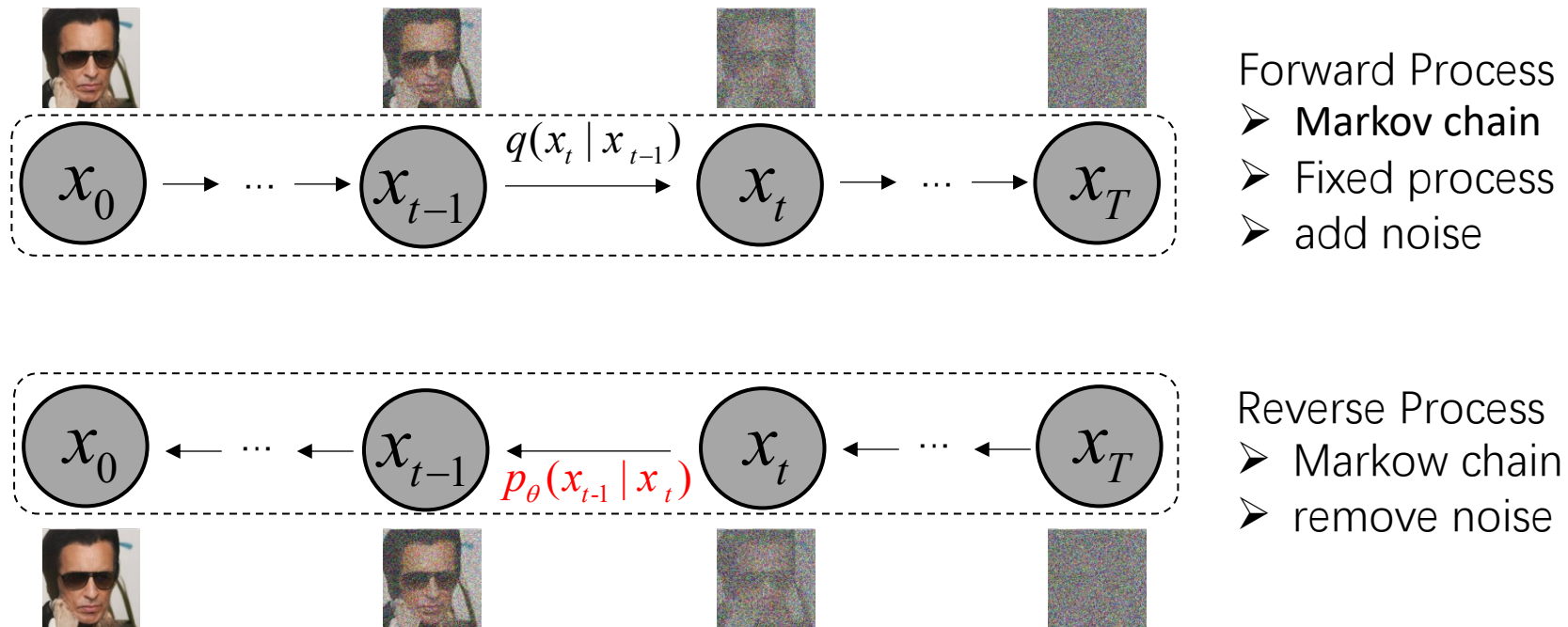


Can deterministic models handle real-world **complexity**?

What about ...



Denoising diffusion probabilistic models (DDPM) - *A powerful generative model*



1. How to generalize **DDPM** to stochastic STG forecasting?
2. How to capture the **ST-correlation** in p_θ ?
3. How to make it **efficient** in the reverse process?

Theory-Driven - *Uncertainty*

Motivation

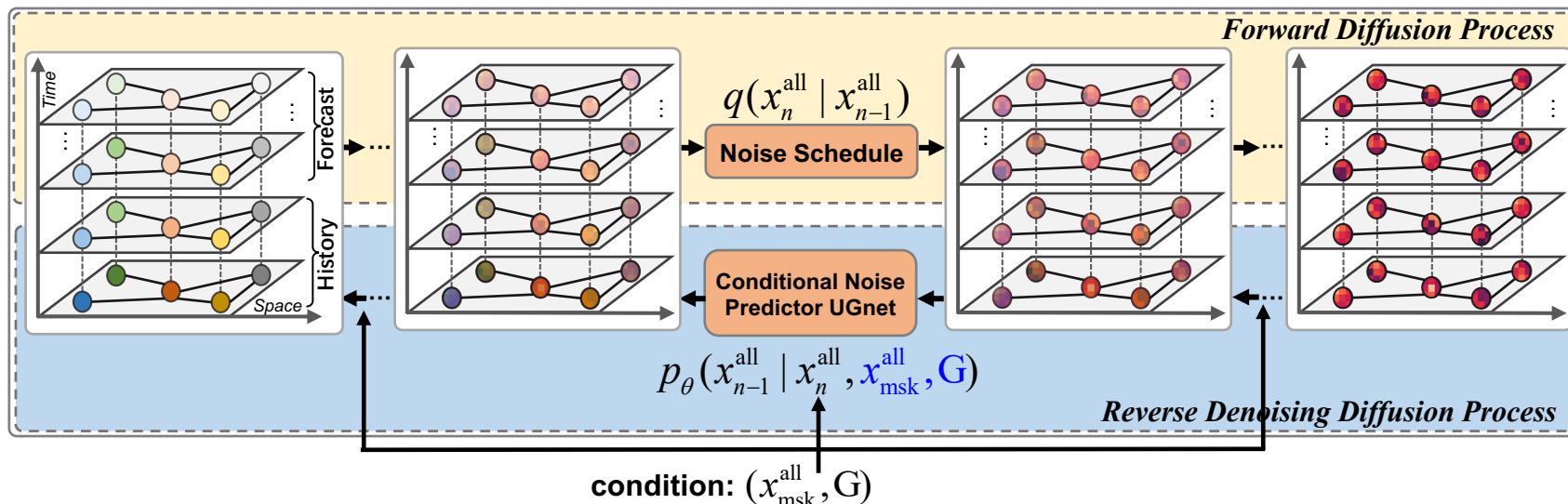
Background – DDPM

Key Challenges

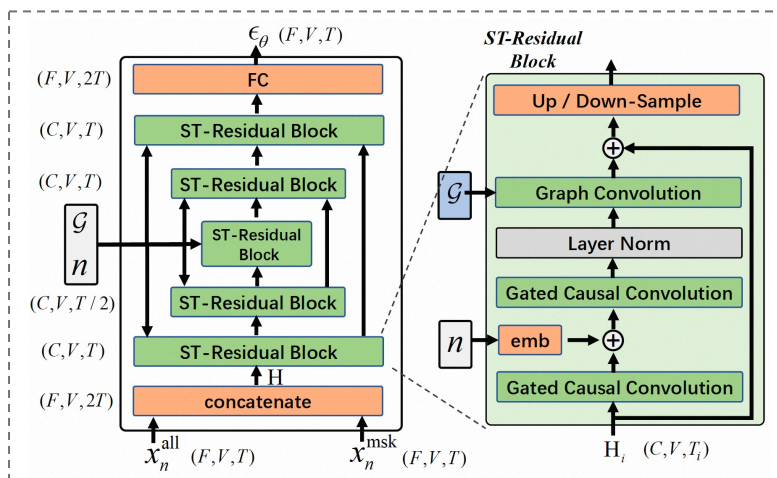
Our Solution - DiffSTG

Experiments Results

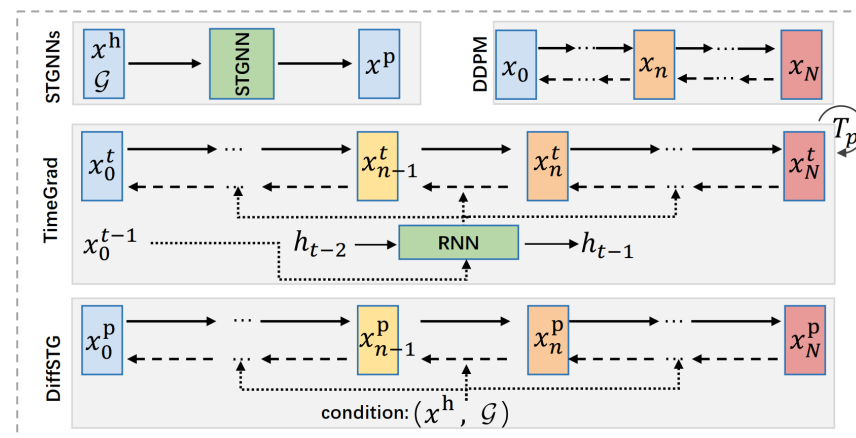
1. Generalize DDPM to STG forecasting - **DiffSTG**



2. Capture the ST-correlation in p_θ - **UGnet**



3. Efficient in the reverse process – **One diffusion loop for all predictions + skip diffusion steps**



Datasets

Dataset	Nodes	F	Data Type	Time interval	#Samples
PEMS08	170	1	Traffic flow	5 minutes	17,856
AIR-BJ	34	1	PM _{2.5}	1 hour	8,760
AIR-GZ	41	1	PM _{2.5}	1 hour	8,760

Metrics

$$\text{CRPS}(F, x) = \int_{\mathbb{R}} (F(z) - \mathbb{I}\{x \leq z\})^2 dz,$$

$$\text{MAE}(Y, \hat{Y}) = \frac{1}{|Y|} \sum_{i=1}^{|Y|} |Y_i - \hat{Y}_i|,$$

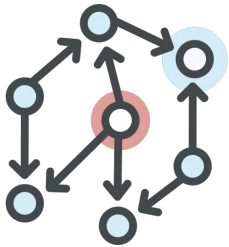
Results

Method	AIR-BJ			AIR-GZ			PEMS08		
	MAE	RMSE	CRPS	MAE	RMSE	CRPS	MAE	RMSE	CRPS
Latent ODE [30]	20.61	32.27	0.47	12.92	18.76	0.30	26.05	39.50	0.11
DeepAR [31]	20.15	32.09	0.37	11.77	17.45	<u>0.23</u>	21.56	33.37	<u>0.07</u>
CSDI [37]	26.52	40.33	0.50	13.75	19.40	0.28	32.11	47.40	0.11
TimeGrad [26]	<u>18.64</u>	<u>31.86</u>	<u>0.36</u>	12.36	18.15	0.25	24.46	38.06	0.09
MC Dropout [44]	20.80	40.54	0.45	<u>11.12</u>	<u>17.07</u>	0.25	<u>19.01</u>	<u>29.35</u>	0.07
DiffSTG (ours)	17.88	29.60	0.34	10.95	16.66	0.22	18.60	28.20	0.06
Error reduction	-4.1%	-7.1%	-5.6%	-1.5%	-2.4%	-4.3%	-2.2%	-3.9%	-14.3%



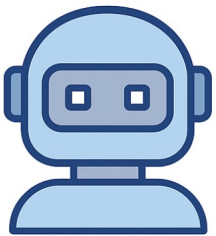
Background

- What makes cities predictable?
- Spatio-Temporal (ST) Data & Properties



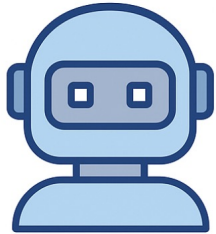
Spatio-Temporal Graph Forecasting

- What is Spatio-Temporal Graph (STG)?
- What is STG forecasting?
- How we do it?
 - Application-Driven (Air Quality, Traffic, Parking)
 - Theory-Driven (Causality, Uncertainty)



Beyond Prediction: What's Next?

- LLMs-powered Agents & Causal Urban Insight



Beyond Prediction: What's Next?

- LLMs-powered Agents & Causal Urban Insight

What next?



We don't just want to predict cities — we want to understand them.



Beyond Prediction:

Toward a More Intelligent
and Accessible Urban Causal Analysis



*Causal or
coincidence?*



*Do **food court closures** cause more people to **order delivery**?*



*Causal or
coincidence?*



*Do **food court closures** cause more people to **order delivery**?*

Confounding factors: time of year, ongoing promotions, income levels, weather

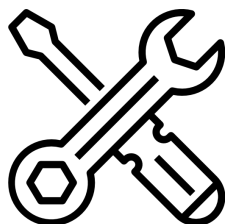
This is where **causal inference** tools become necessary.

$$\underbrace{y_i}_{\text{Outcome}} = \underbrace{T_i}_{\text{Treatment}} \underbrace{\beta_i}_{\text{Causal Effect}} + \underbrace{X_i}_{\text{Control Variable}} \underbrace{\delta_i}_{\text{Unobserved Confounders}} + \underbrace{U_i}_{\text{Unobserved Confounders}} + \underbrace{e_i}_{\text{Stochastic Error}}$$

Urban Causal Research



Urban
Phenomenon



Causal Inference



*Causal or
coincidence?*



Do *food court closures* cause more people to *order delivery*?

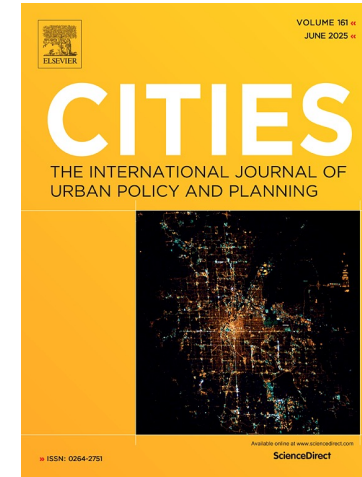
Confounding factors: time of year, ongoing promotions, income levels, weather

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$$\begin{array}{ccccccc}
 & \text{Treatment} & & \text{Unobserved} & & & \\
 & & & \text{Confounders} & & & \\
 \boxed{y_i} = & \boxed{T_i \beta_i} + & \boxed{X_i \delta_i} + & \boxed{U_i} + & \boxed{e_i} & \text{Stochastic} \\
 \text{Outcome} & \text{Causal} & \text{Control} & & & \text{Error} \\
 & \text{Effect} & \text{Variable} & & &
 \end{array}$$

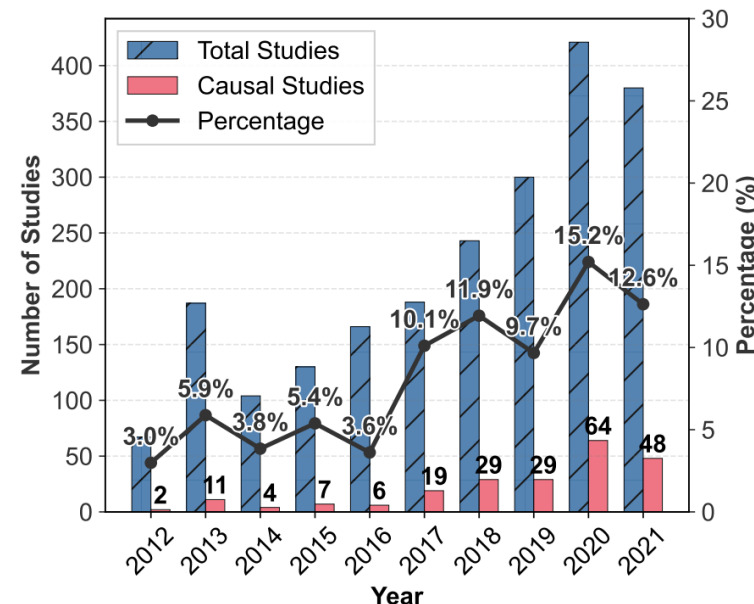
Systematic Review

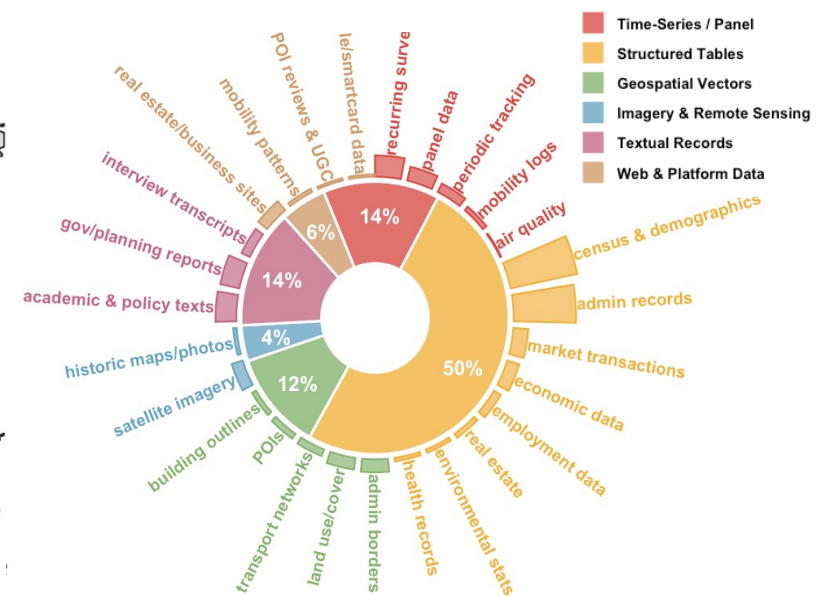
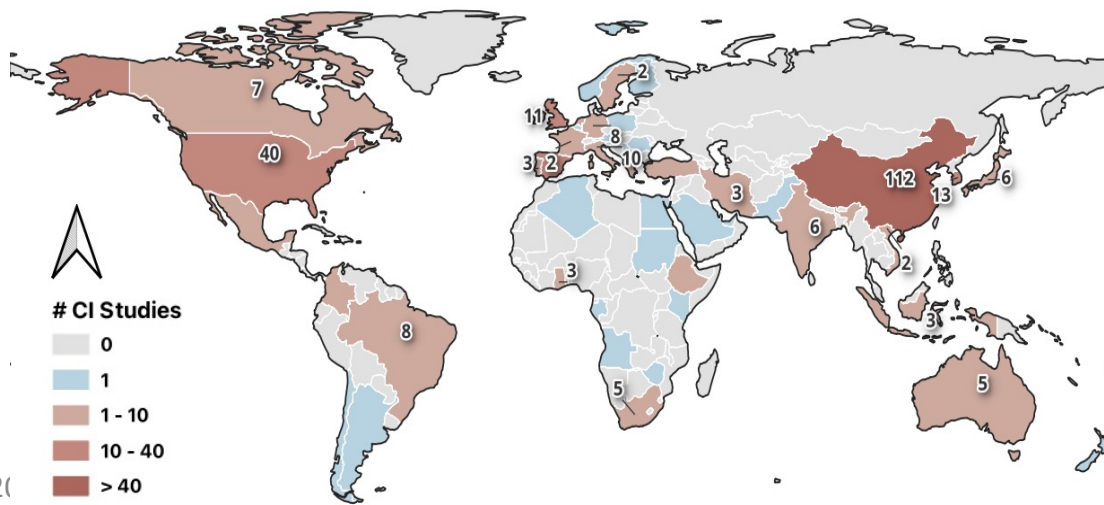
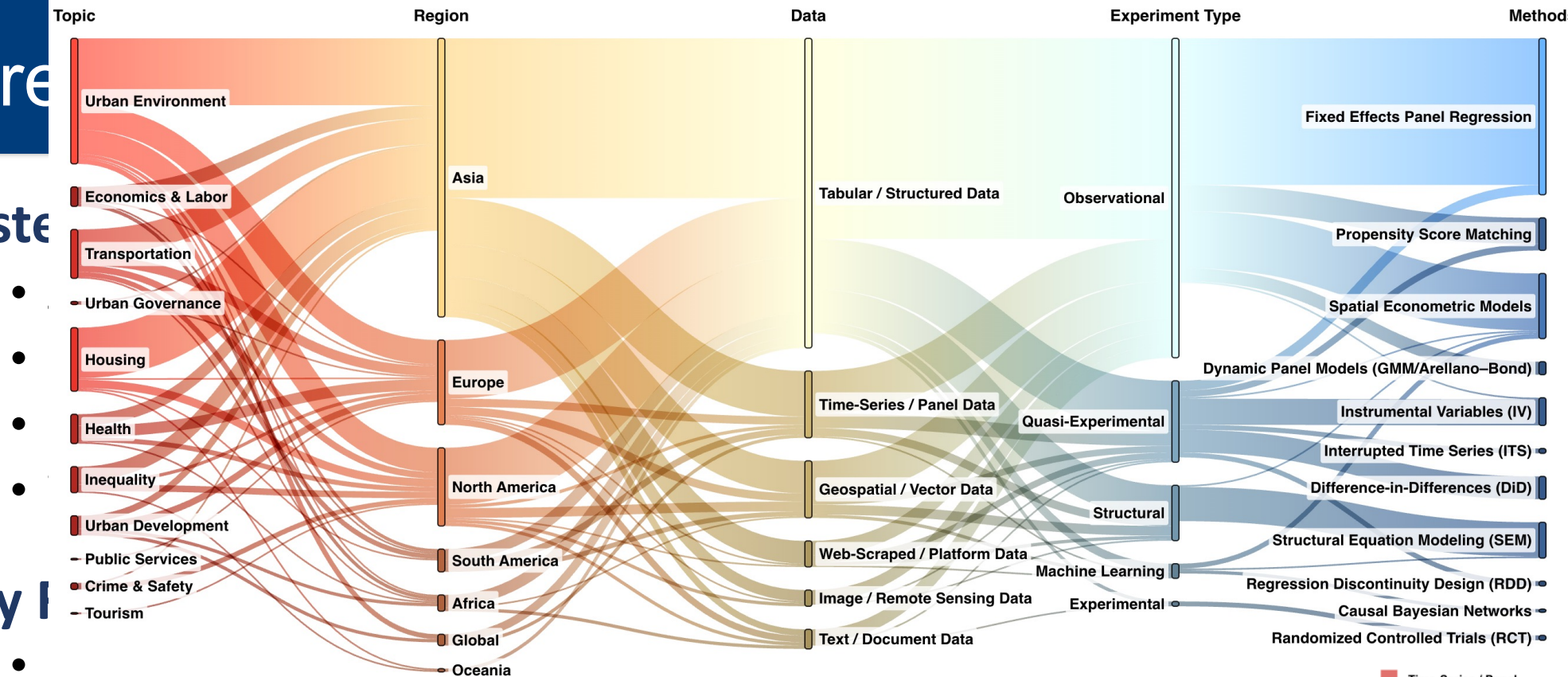
- Journal: *Cities*
- # total urban-related papers: **2,428** articles
- # total causal inference papers: **19** articles
- Timespan: **2012–2021**



Key Findings:

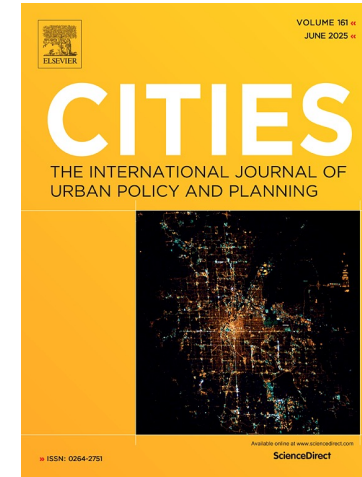
- **Trend:** growing adoption





Systematic Review

- Journal: *Cities*
- # total urban-related papers: **2,428** articles
- # total causal inference papers: **19** articles
- Timespan: **2012–2021**



Key Findings:

- **Trend:** growing adoption
- **Region:** geographical imbalances
- **Data:** a heavy reliance on structured data
- **Method:** limited methodological diversity
- **Code:** poor reproducibility



Manual Workflow Limitizations



Phase 1 Generating Hypotheses

- Challenges in timely and novel question formulation
- Subjectivity in hypothesis formation



Phase 2 Assembling Urban Data

- Multimodal, multisource, and multiresolution data complexity
- Repetitive manual data integration and cleaning



Phase 3 Executing CI Experiments

- Difficulty selecting the most suitable causal inference methods
- Ensuring methodological rigor



Phase 4 Drawing Policy Insights

- Translating findings into generalizable, policy-relevant insights

Manual Urban Causal Research



Manual Workflow Limitizations



Phase 1

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Phase 4

Drawing Policy
Insights

- Translating findings into generalizable, policy-relevant insights

Multiple Sources



Geographical Data

POI/AOI; Satellite Image; Street-view Image



Traffic Data

Trajectory; Road Network; Traffic Flow; Logistic



Social Media Data

Geo-textual Data; Geo-tagged Photo;
Geo-tagged Video



Demographic Data

Population; Crime Data; Land Use Data



Environment Data

Meteorological Data; Greenery Data;
Air Quality Data

Multiple Modalities

Spatio-temporal Modality

POI; GPS Point; Time series; Trajectory...



Visual Modality



Satellite Image



Street-view Image

Textual Modality

Social Media Text:

Users' comment : Arrived at an impressive park surrounded with...

Geo-information Text:

City A is located at the subtropics with ...

Other Modalities

Audio; Video; Hyper-spectrum..



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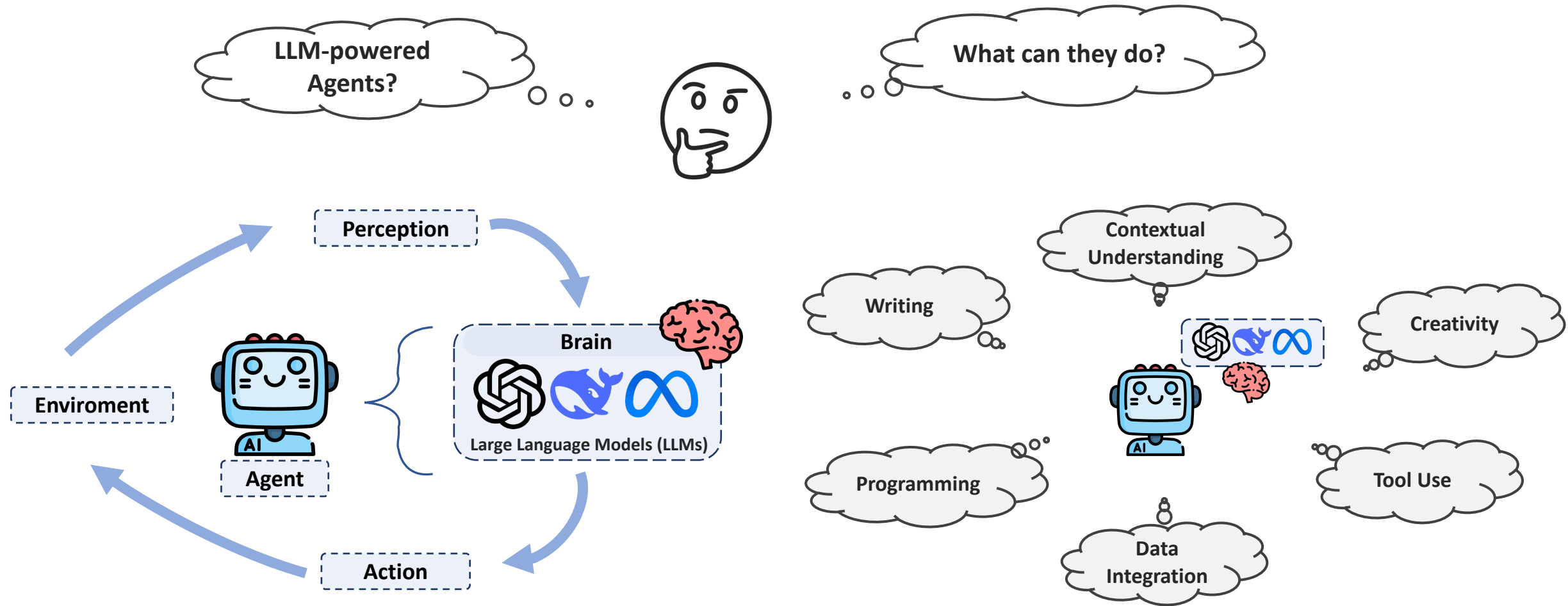


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Can LLMs Help?

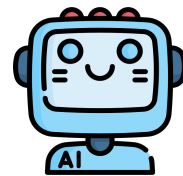
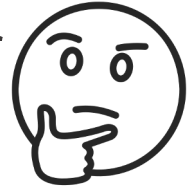
LLMs & Agents



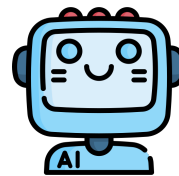
What happens when putting **multiple agents** together in an environment?



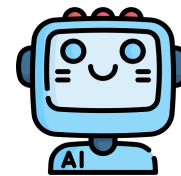
What if we created **a team of agents** to simulate an **urban research lab**?



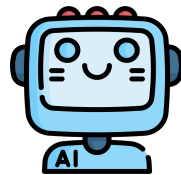
Agent 1



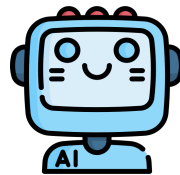
Agent 2



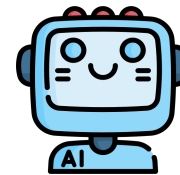
Agent 3



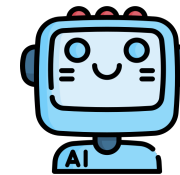
Agent 4



Agent 5

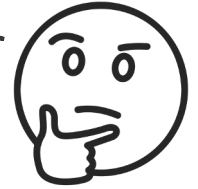


Agent 6



Agent 7

What if we created **a team of agents** to simulate an **urban research lab**?



Urban Scientist



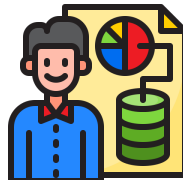
Data Scientist



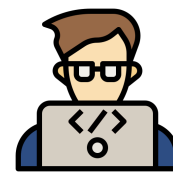
Validator



Reader



Data Engineer



Experimenter



Writer



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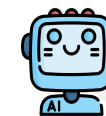
- Difficulty selecting the most suitable causal inference methods
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Phase 4
Drawing Policy Insights

- Translating findings into generalizable, policy-relevant insights

Motivate



LLMs/MLLMs' Abilities

Knowledge reasoning

Contextual Understanding



Creativity



Writing



Procedural execution

Data Integration



Programming

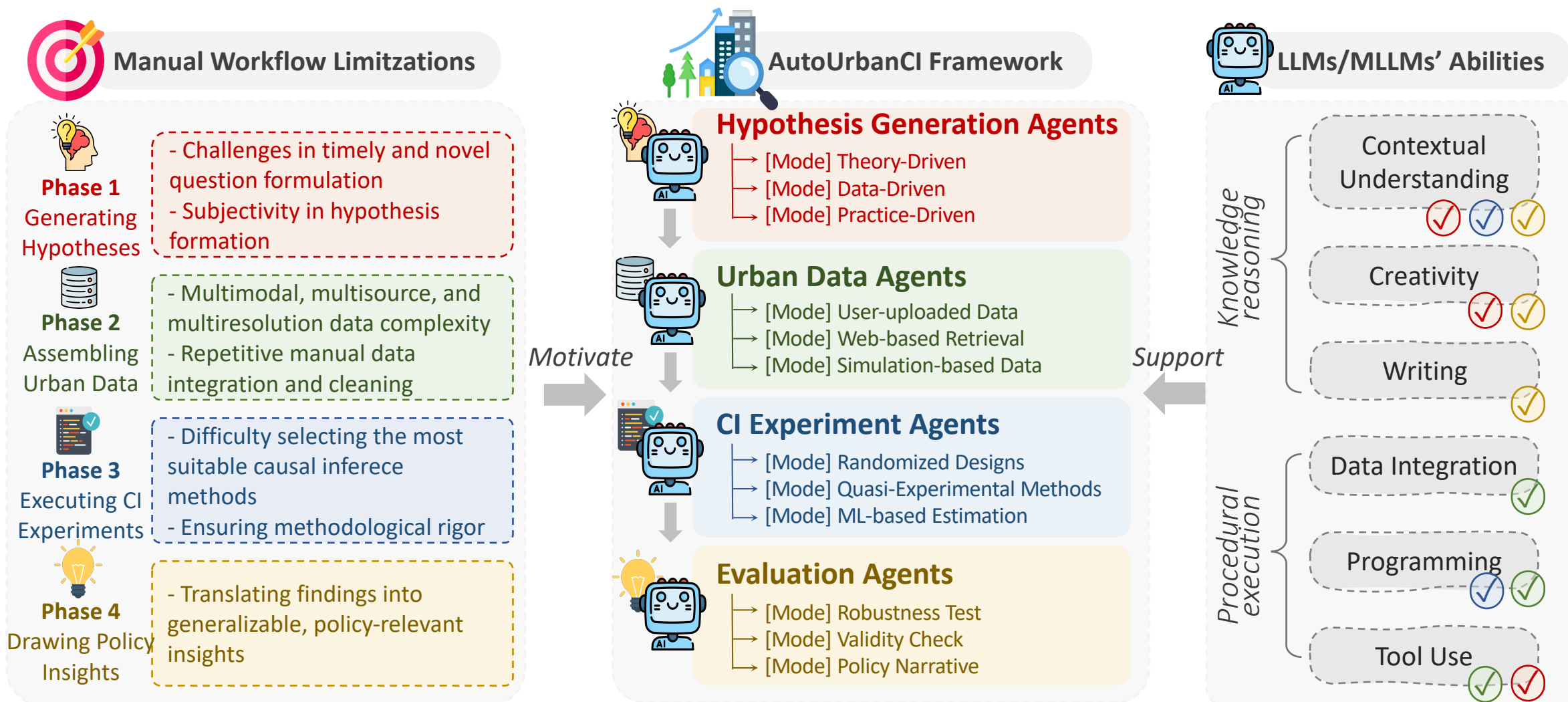


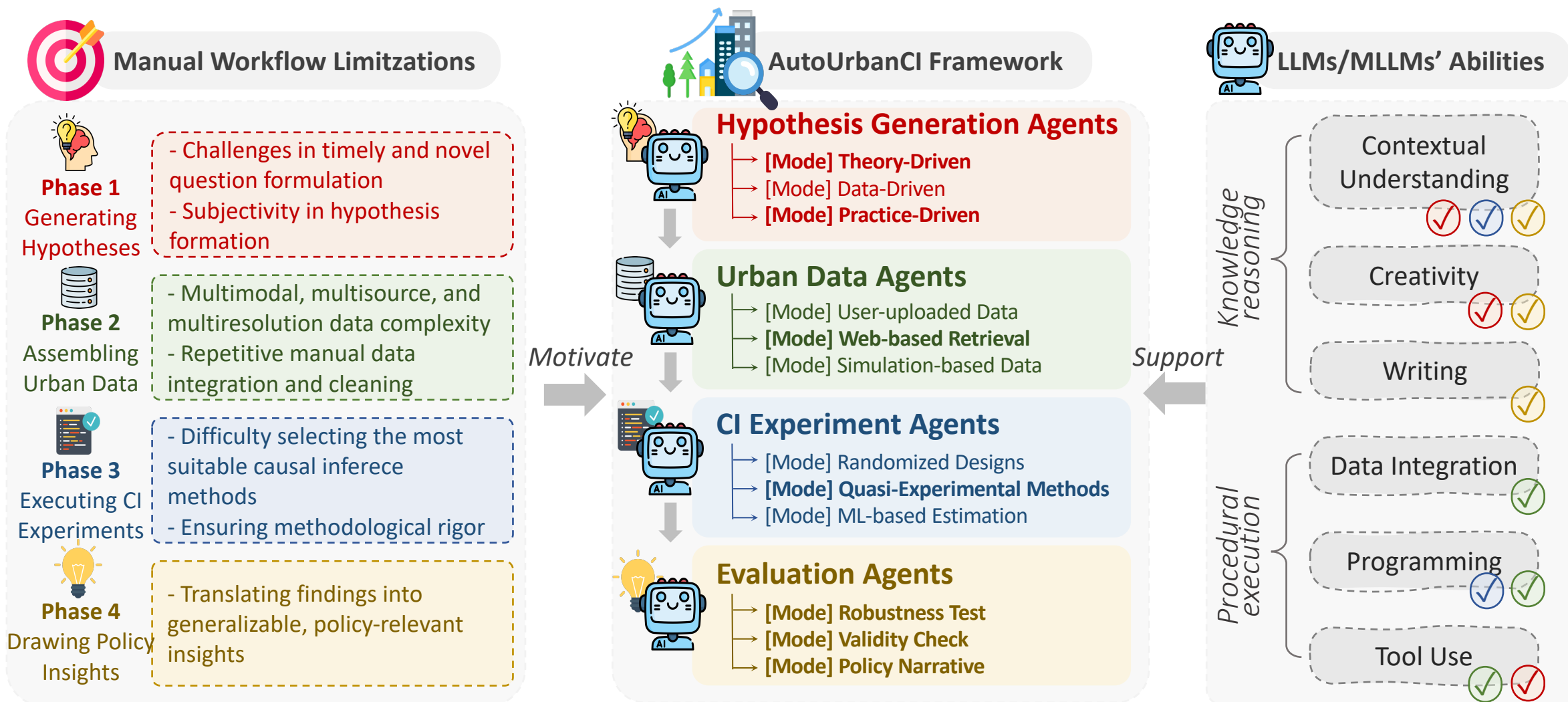
Tool Use



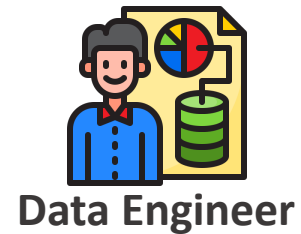
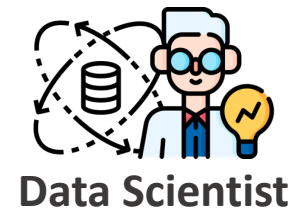
Support





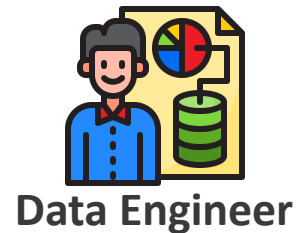
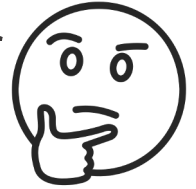


A Example Use Case



A Example Use Case

What if we created **a team of agents** to simulate an **urban research lab**?

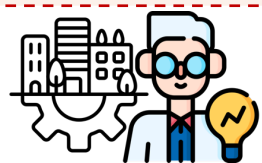


A Example Use Case

Hypothesis Generation

Validate Hypothesis & Define Data Needs

Adjust hypothesis based on policy needs, define data requirements



Urban Scientist



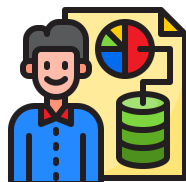
Refine Hypothesis



Reader

Hypothesis: "In NYC, congestion pricing reduces average commute times less significantly in low-income neighborhoods than in higher-income areas."

Required Data: Subway/bus logs, taxi trips, street view images, social media posts, and census income data.



Data Engineer



Experimenter



Data Scientist

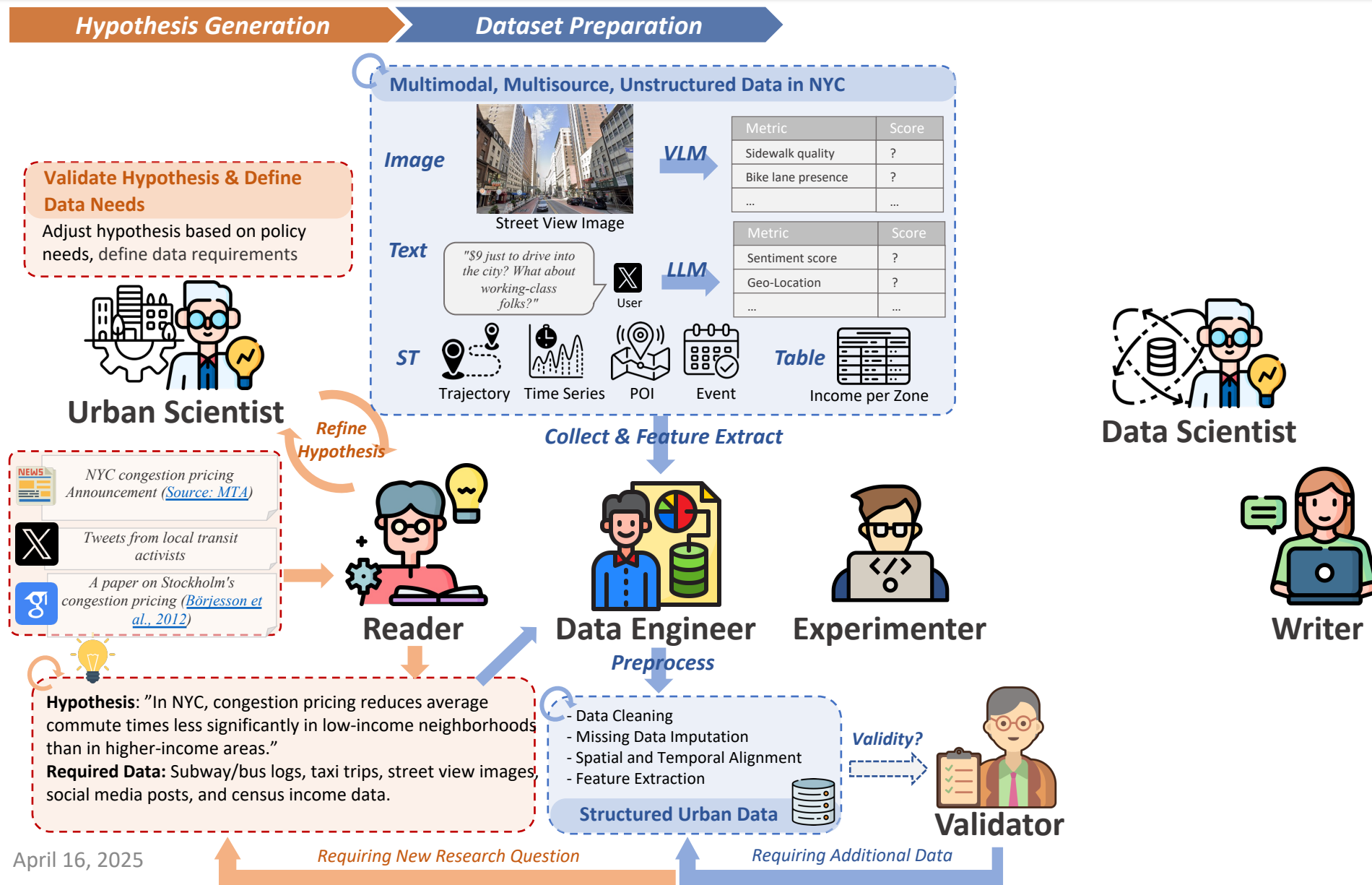


Writer

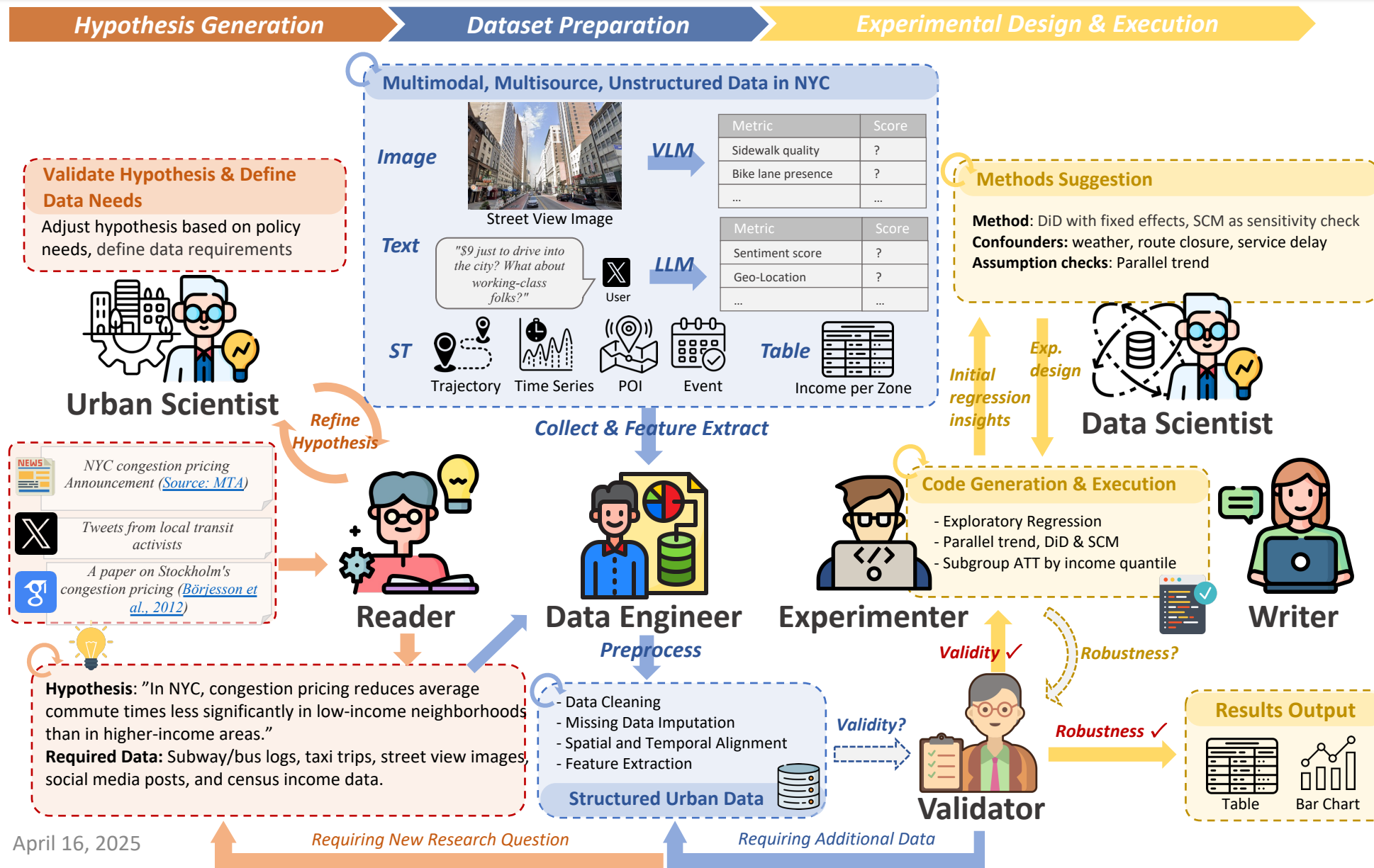


Validator

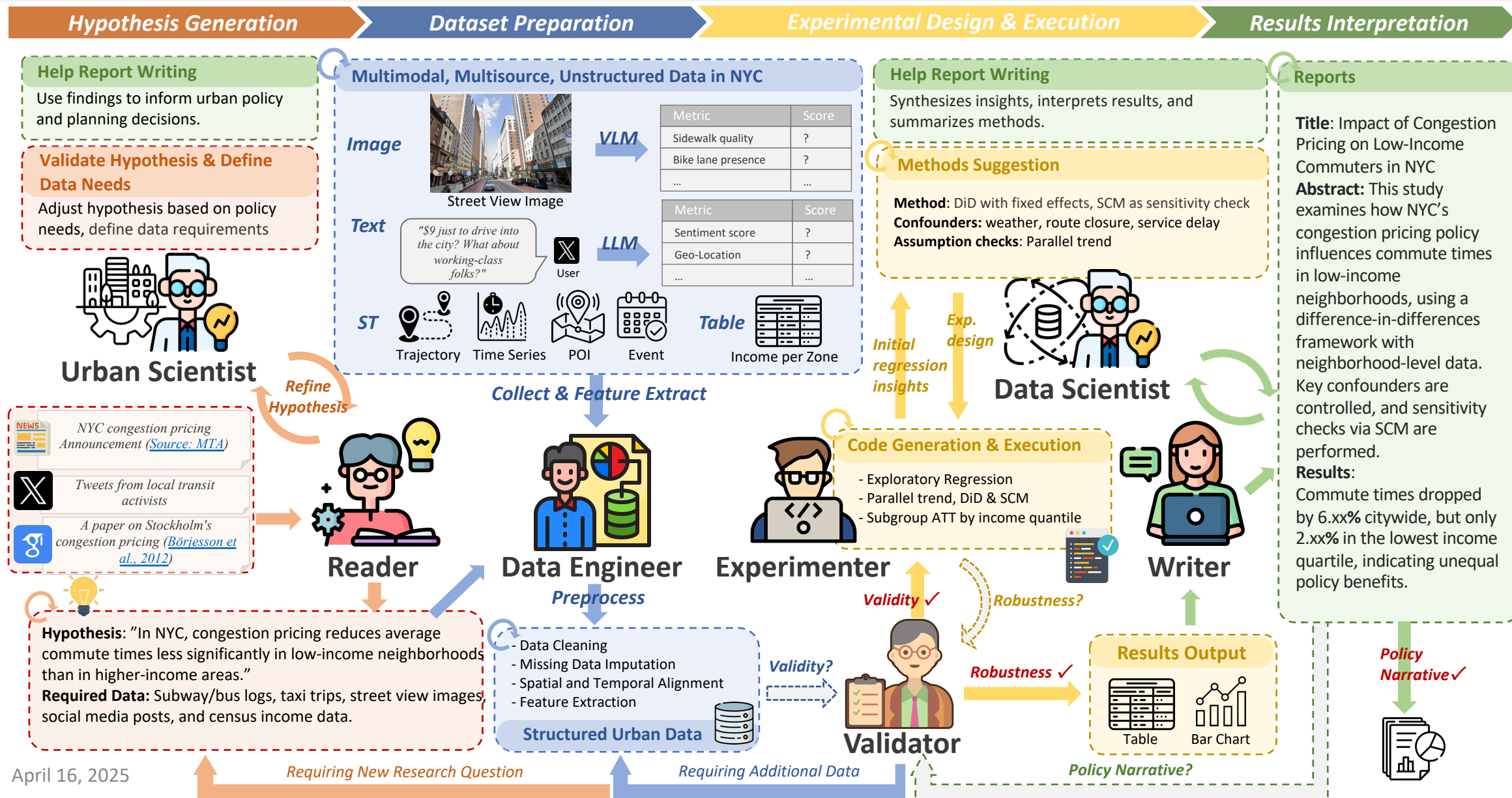
A Example Use Case



A Example Use Case



A Example Use Case

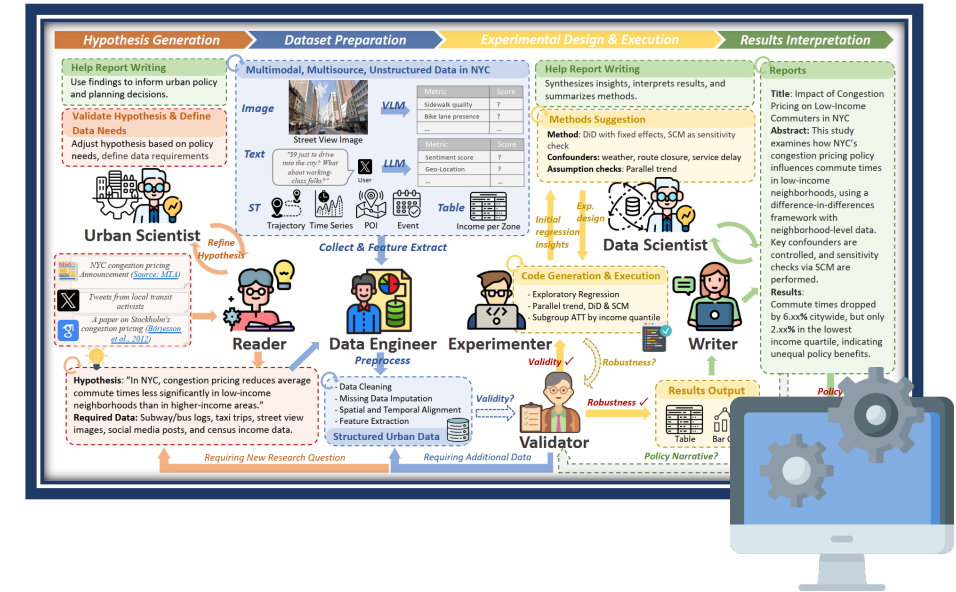


For *Urban Researchers*:

- Assists and accelerates causal research

For *Policy Makers*

- Enhances evidence-based urban policy



For *Urban Researchers*:

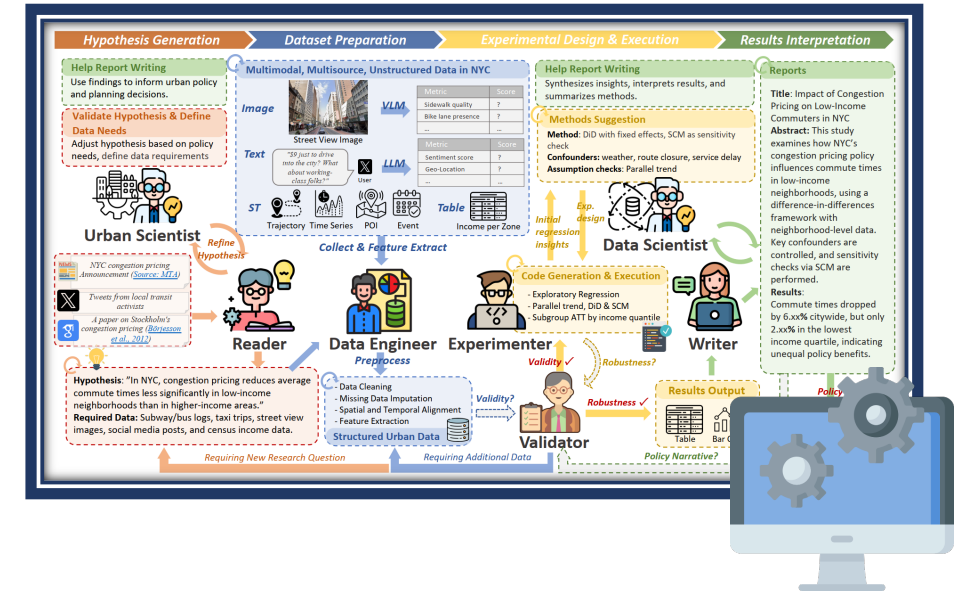
- Assists and accelerates causal research

For *Policy Makers*

- Enhances evidence-based urban policy

For the *Public*:

- Lowers the barriers for citizens, journalists, and grassroots organizations to explore urban issues



*“Cities have the capability of providing something for **everybody**, only because, and only when, they are created by **everybody**.”*

— Jane Jacobs





AI won't replace urban researchers,
but it can *assist* their thinking, *accelerate* their work,
expand the horizons of urban science, and make urban
insights a *shared power*.
That's the *future* we're building toward.

- Cities are predictable.
 - *STG forecasting* gives us powerful tools to model these signals in domains like air quality and mobility.
 - *Application-driven* modeling is tailored to the properties of specific data (e.g., air quality, traffic),
 - *Theory-driven* approaches incorporate causal lens and uncertainty modeling for deeper insight.
- Towards prediction, what next?
 - Toward a more *intelligent* and *accessible* urban causal analysis
- Large Language Models unlock new possibilities.
 - Not to replace urban scientists, but *assist* their thinking, *accelerate* their work, *expand* the horizons of urban science, and make urban insights a *shared power*.

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Thanks!